

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

**Analyzing and Advancing Unpaired PET to CT Image Translation:
Improvements through Enhanced CycleGAN and CUT**

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Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Tanvir Hossain Saikat**, **Arafat Bin Amin Drubo**, and **Najmus Sakib Arko** under the supervision of **Tareque Mohmud Chowdhury**, Assistant Professor, Department of Computer Science and Engineering and co-supervision of **Sabrina Islam**, Lecturer, Department of Computer Science and Engineering and **Farzana Tabassum**, Lecturer, Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others have been acknowledged in the text and a list of references is given.

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Dedicated to Our Parents

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Abstract

Medical imaging plays a crucial role in healthcare, with PET and CT scans providing complementary information about metabolic activity and detailed anatomical structures, respectively. However, the reliance on paired PET-CT data poses challenges in terms of data availability, cost, and clinical variability. This research addresses the unpaired translation of PET images to CT images. Motivated by the potential for improved decision-making and the cost-effective utilization of unpaired data, this work aims to overcome the limitations of current medical imaging approaches & to address the effectiveness of this task. The study explores the use of unpaired image to image translation models for the given task including - CycleGAN, CUT, ,UNIT, MUNIT, SCAM-GAN etc & proposed improvements with the current state of the art result for this specific task. The literature review highlights the effectiveness of these models in various domains, emphasizing their loss functions, including adversarial, identity, cycle consistency & perceptual losses, adding attention mechanism and so on. To enhance the model's capabilities, the research incorporates attention mechanisms aims to capture dependencies in images. The study evaluates the proposed methodology's accuracy and applicability by comparing results with existing models. The research team aims to demonstrate the feasibility of solving this task by evaluating different state of the art unpaired models & proposes incorporations which gives better accuracy than the current models.

Keywords — unpaired image to image translation problem, image processing, GAN, cycleGAN, attention mechanism, MS-SSIM, CUT, spatial attention, self attention, contrastive learning, perceptual loss, VGG-16, PET to CT, hinge loss objective, MUNIT, medicale image translation, SSIM

Chapter 1

Introduction

1.1 Overview

Image translation is a computational process that involves transforming an image from one domain to another while preserving relevant visual content and characteristics. This transformative task is crucial in various fields, such as computer vision and medical imaging, where different imaging modalities capture distinct aspects of the same underlying scene. The objective is to enable seamless translation between domains with disparate features, facilitating tasks like style transfer, object recognition, and medical diagnosis. Recent advancements in deep learning, particularly with generative adversarial networks [6] (GANs) and their variants like cycleGAN [36] & CUT [21], have significantly improved the ability to perform image translation even in the absence of paired data. This not only addresses the challenge of acquiring matched samples for training but also opens avenues for innovative applications, ranging from artistic image synthesis to the integration of diverse medical imaging modalities for comprehensive analysis and diagnosis.

Positron Emission Tomography (PET) and Computed Tomography (CT) are two distinct yet complementary imaging modalities widely used in the field of medical diagnostics. PET imaging focuses on detecting and visualizing metabolic activity within the body by tracing the distribution of radiotracers that emit positrons. This provides valuable information about cellular function and physiological processes, aiding in the identification and characterization of various diseases, including cancer. On the other hand, CT imaging utilizes X-rays to create detailed cross-sectional images of the body's internal structures, offering precise anatomical information. CT scans are particularly adept at visualizing bones, soft tissues, and organs with high

spatial resolution. The integration of PET and CT scans, commonly referred to as PET-CT imaging, combines the strengths of both modalities. This hybrid approach allows for a comprehensive assessment of both metabolic activity and anatomical structures in a single examination, enabling more accurate diagnoses, treatment planning, and monitoring of various medical conditions.

In the realm of machine learning and image processing, the concepts of paired and unpaired data play pivotal roles in shaping the effectiveness of algorithms. Paired data refers to a set of observations or instances in which each element in one dataset has a direct corresponding counterpart in another dataset. This one-to-one correspondence is particularly valuable for supervised learning tasks, as it facilitates the training of models by establishing clear associations between input and output pairs. However, acquiring paired data can be resource-intensive and challenging in various domains. In contrast, unpaired data comprises instances from two datasets that lack direct one-to-one correlations. Unpaired data is often more readily available but presents challenges in terms of training models, as the absence of explicit correspondences can make it harder for algorithms to learn meaningful relationships. Techniques such as cycleGAN or CUT, which excels at unpaired image-to-image translation, have been developed to address the limitations associated with the scarcity and cost of acquiring paired datasets, enabling the exploration of diverse applications in fields like medical imaging and computer vision.

1.2 Research Challenges

The prominent research challenges that are present in the realm of unpaired image to image translation are,

- Unpaired image translation involves learning mappings between two domains without direct correspondences between individual images. Aligning features and styles across domains without paired samples is a significant challenge.
- Addressing the domain gap between source and target domains is a fundamental challenge. Distributions of images in different domains may vary, making it difficult for models to generalize well to unseen data.
- Ensuring that the content and semantics of the images are preserved during translation is crucial. Some methods may struggle with maintaining the essential information while changing the style or domain.
- Developing reliable evaluation metrics for unpaired image translation is non-

trivial. Traditional metrics like PSNR and SSIM may not capture perceptual differences well. Human perceptual studies or domain-specific metrics are often needed.

- Achieving fine-grained control over the translation process, especially when dealing with intricate details, textures, or small objects in images, remains challenging.
- Unpaired translation models often require large amounts of diverse data for effective training. Generalizing well to diverse and unseen data is a challenge, especially when dealing with limited amounts of training data.
- Training generative models, particularly those using adversarial networks, can be challenging due to issues like mode collapse and training instability. Ensuring stability during the training process is an ongoing concern.
- Extracting meaningful and disentangled features from unpaired data is a challenging problem. Learning representations that capture both shared and domain-specific information is crucial for effective translation.
- Combining CT with PET can lead to better medical decision making but finding both PET, CT of a single patient at the same time is rare, costly & time consuming
- Only one work has been done before on this specific task of unpaired PET to CT translation

1.3 Thesis Contributions

To account for the aforementioned limitations, we present our thesis work that yields the following contributions —

- As there has been only one work done, we analyze the task’s result with different state-of-the art unpaired models & compile them to measure their significance
- We propose better frameworks for unpaired PET to CT image translation using cycleGAN & CUT with attention mechanism & structure consistent loss.
- We also generalized our model so that it can capture the underlying representation between any two domain where the geometric changes or structural dissimilarities is very large & paired data is not available

cycleGAN & CUT are currently two of the most popular adversarial models due to its effectiveness in achieving state-of-the-art results on a variety of unpaired image to image translation.

1.4 Motivation

The motivation behind this research lies in the imperative need to enhance decision-making in healthcare by bridging the information gap between PET and CT imaging modalities. While PET emphasizes metabolic activity and CT provides detailed anatomical structures, the current reliance on paired PET-CT data poses substantial challenges in terms of data availability, cost, and clinical variability. By tackling the translation of PET images to CT images in an unpaired manner, the research aims to not only streamline the imaging process, making it more cost-effective and accessible, but also to address the complexities introduced by variations in imaging conditions and the time-consuming nature of collecting paired datasets. The broader goal is to establish a more generalized and robust approach applicable to scenarios where structural dissimilarities are significant, and obtaining paired data is impractical or prohibitively expensive.

1.5 Problem Formulation

Given a PET image P we need to synthesize a corresponding paired CT image C

1.6 Thesis Layout

The thesis is thoughtfully structured, adhering to a well-crafted layout that encompasses the following 6 chapters — In Chapter-1, we set the context for the reader by introducing the research problem and delineating the challenges encountered in this domain of scholarly inquiry. We formulate the research problem in precise terms, pristinely list down the contributions of this thesis work, and write about the circumstances that inspired us to pursue the topic in this chapter. This thesis layout section is a constituent of the chapter that aids in the readability of the thesis book by recapitulating a cohesive narrative of the work. In Chapter-2, we critically examine the existing scholarly works, theories, and empirical studies related to the research topic, establishing a strong theoretical foundation for the study. We provide a proper taxonomy and chronological development of the models used in the

existing literature in this chapter and perform comparative analyses among them. In Chapter-3, we outline our proposed methodology with the aid of intuitive diagrams and explain the underlying mechanisms of the components that are present within the architecture. Chapter-4 elaborates on the experimental setup or procedures undertaken to collect empirical data. We provide a detailed statistical analysis of our dataset and delineate the criteria that we have taken into account whilst constructing the dataset. We also mention the models that we use as baselines and the intricate details pertaining to the training process in this chapter. Chapter-5 presents an in-depth analysis and interpretation of the obtained results along with a critical discussion of their implications and relevance. We also bring to light some of the weaknesses of the existing models and provide a comprehensive ablation study to convey a deeper understanding of the pipeline’s working process to the reader in this chapter. The epilogue of our thesis work, Chapter-6, provides a concise summary of our research findings and insights into potential future directions for further exploration towards the apotheosis of this research domain.

Chapter 2

Literature Review

2.1 Image Translation in Medical Imaging

In the realm of medical imaging, the fusion of information from different imaging modalities holds immense promise for advancing diagnostics and treatment planning. Image translation, a transformative concept in this context, refers to the process of converting images from one modality to another, thereby combining distinct but complementary information. Notably, this integration is particularly critical in the case of Positron Emission Tomography (PET) and Computed Tomography (CT), where PET emphasizes metabolic activity, while CT provides detailed anatomical structures[12]. The amalgamation of these modalities can significantly enhance clinical decision-making, offering a comprehensive understanding of both functional and structural aspects. Traditionally, medical imaging has heavily relied on paired datasets, where corresponding images from both modalities are acquired simultaneously for the same patient. However, this dependence on paired data poses challenges in terms of accessibility, cost, and the intricate task of establishing a one-to-one correspondence between PET and CT images. The emergence of unpaired image translation techniques, such as CycleGAN[36], & CUT presents a compelling solution to overcome these challenges by enabling the translation of PET images to their corresponding CT counterparts without the need for paired datasets.

2.2 Unpaired Image to Image translation

Unpaired image-to-image translation is a fascinating sub-field of computer vision tackling the challenge of transforming images from one domain to another without

the requirement of paired training data. This means you can, for example, convert a black and white photo to its colorized version, alter the season in a landscape image, or even translate artistic styles like from Van Gogh to Monet, all without needing perfectly aligned examples for each transformation.

This task poses unique challenges as the model lacks explicit guidance on how the content and style should change across domains. Researchers have devised ingenious solutions, often employing generative adversarial networks (GANs)

Unpaired image-to-image translation opens doors to numerous applications, from artistic exploration and fashion design to medical image enhancement and historical document restoration. As research progresses, the boundaries between domains will continue to blur, paving the way for boundless creative possibilities and real-world transformations.

2.2.1 CycleGAN

Presented in details in 2.4

2.2.2 CUT

Presented in details in 2.5

2.2.3 UVCGAN

The paper by Torbunov et al. (2022)[lit3](#) presents the model UVCGAN: UNet Vision Transformer Cycle-Consistent GAN, where the integration of UNet, Vision Transformers (ViTs), and cycle-consistent GANs is proposed for boosted performance of unpaired image-to-image translation tasks.

The basic idea of UVCGAN is to combine the hierarchical feature extraction capability of UNet, the powerful self-attention mechanism of Vision Transformers, and the cycle consistency principle of GANs. Given the effective encoder-decoder structure of UNet in handling local and global features of images, Vision Transformers are incorporated to enhance feature representation originating from the modeling of long-range dependencies through self-attention. Then, the cycle-consistency loss enforces that a transformed image from one domain to the other and back is the same image as in the original, hence the maintenance of one-to-one correspondence. This property is crucial for scientific uses.

The novelty of UVCGAN is to combine Vision Transformers with the cycle-consistent GAN framework. Other non-Vision Transformer models, such as those applied by CycleGAN, are based on convolutional neural networks for feature extraction. This means that UVCGAN should observe a more significant benefit from upgrading to Vision Transformers because of their advanced abilities of representation in features, which allow one to capture small details and global context more effectively, hence producing the best results.

The strengths of UVCGAN lie in the ability to make a high-quality translation of images, effectively uniting all the strengths of both UNet, Vision Transformer, and GANs. The self-attention mechanism of Vision Transformers allows for handling complex transformations and capturing detailed features in the images. The cycle-consistency loss should ensure real-looking, plausible outputs in translating the image content. Moreover, UVCGAN can work with unpaired data that are based on a lot of real-world applications where paired datasets are not available.

But, at the same time, UVCGAN has its drawbacks. Integrating the architecture of Vision Transformer into the model makes the model complex, incurring a higher computational need, which in turn prolongs the training time and resource consumption. The tuning of the hyperparameters is particularly sensitive to obtain state-of-the-art performance, especially for hyperparameters related to the Vision Transformers, and training becomes a bit tough. Besides, like with other GAN-based models, UVCGAN can either suffer from instability during training or create artifacts improperly trained.

In summary, UVCGAN denotes a giant leap in unpaired image-to-image translation by integrating UNet, Vision Transformers, and Cycle-Consistent GANs. Despite its increased complexity and computational requirements, UVCGAN provides a robust framework for producing high-quality, consistent image translations and is suitable for various applications in image processing and computer vision.

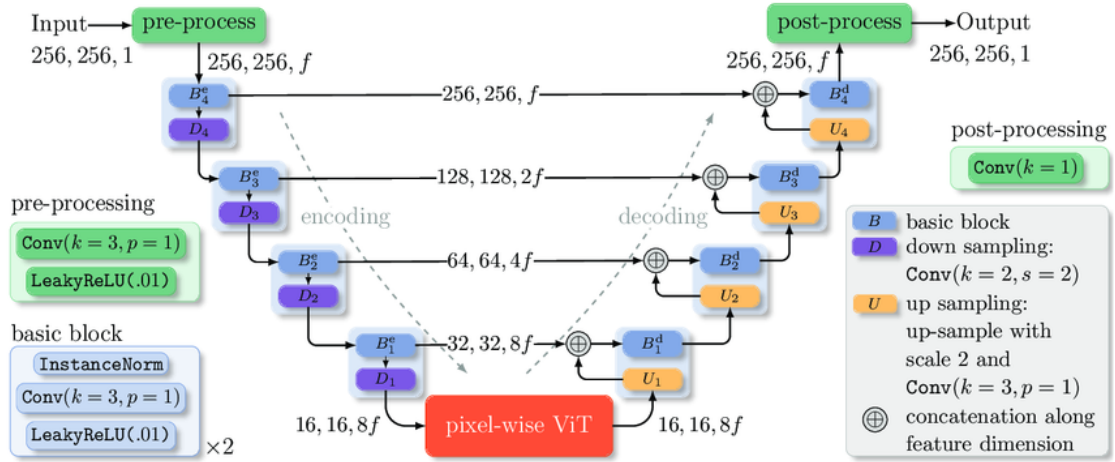


Figure 2.1: UVCGAN Architecture

2.2.4 TransGAN

The paper "TransGAN: Two Pure Transformers Can Make One Strong GAN, and That Can Scale Up"^{lit4} by Jiang et al. (2021) introduces TransGAN or Transformer GAN. This model studies the usability of transformers, initially conceived for natural language processing. Still, now they have been adapted to a broad task list in computer vision as the model to be put in the core architecture for GANs.

The key idea of TransGAN is to replace the convolutions in traditional GANs with transformer modules. In that respect, TransGAN consists of a transformer-based generator with incrementally increased feature resolution and a multi-scale transformer-based discriminator capturing not only semantic contexts but low-level textures. The model also introduces grid self-attention to eliminate memory bottlenecks and, hence scale up to high-resolution image generation.

What distinguishes TransGAN from other GAN models is that it has done away with convolutional layers, which essentially comprise the core of StyleGAN and CycleGAN architectures. Pure transformer architectures will help TransGAN wring more from what makes transformers tick: modeling long-range dependencies, as well as capturing rich contextual information. This work is a pioneering attempt to build nonconvolutional GANs, opening up a new way for GAN research.

Consequently, the strengths of TransGAN lie in its capability to pull off state-of-the-art performance on several benchmarks. For instance, a new inception score of 10.43 and FID of 18.28 are achieved over the STL-10 dataset, outperforming StyleGAN-V2. It further shows competitive performance on CIFAR-10 and CelebA datasets with produced qualitative images having remarkable texture detail. In addition, its

transformer-based architecture opens a new direction regarding investigating more robust and flexible generative models.

Still, with this overall strength, TransGAN is subject to specific weaknesses. Introducing transformers tends to increase the model's complexity, bringing in computational needs for training that are much more laborious than the inference training. The training itself can prove quite unstable, so stabilization calls for the balanced use of multiple techniques combined, like data augmentation, modified normalization, and relative position encoding, among others. Moreover, while transformers can capture long-range dependencies, they may struggle with a few low-level texture details that convolutional networks handle very naturally.

In short, TransGAN is an excellent innovation in a generative model that uses transformers to build convolution-free GAN architecture. Therefore, there are challenges in computational needs for training stability. Still, TransGAN proves the potential of transformers in high-quality image generation and sets a new benchmark for future studies in GANs.

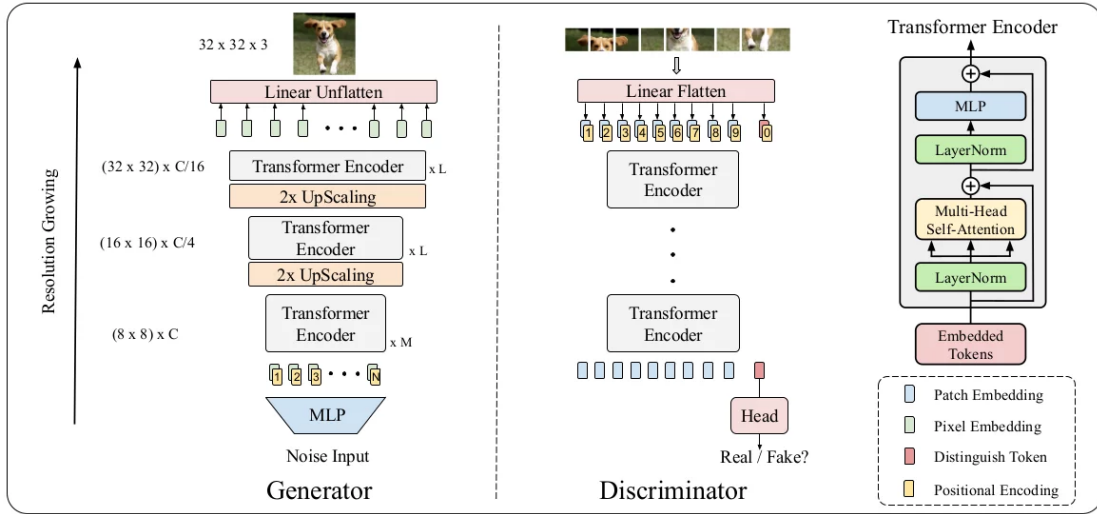


Figure 2.2: TransGAN Architecture

2.2.5 MSPC

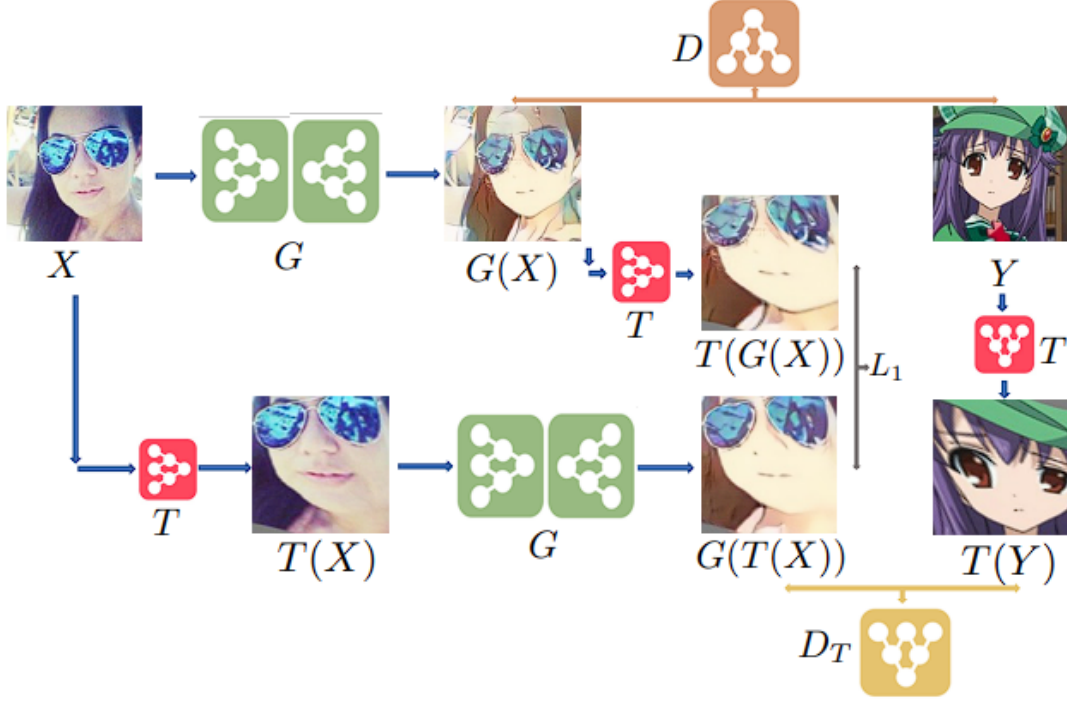


Figure 2.3: MSPC Architecture

Unpaired image-to-image translation recently introduced Maximum Spatial Perturbation Consistency (MSPC) by Xu et al. (2022)^{lit5}. This work, presented at the 2022 CVPR conference, introduces a novel modification to introduce regularization to enhance the quality and robustness of unpaired image-to-image translation tasks.

The basic idea of MSPC is that a spatial perturbation function, T , commutes with a translation operator, G , in formulating

$$T \circ G = G \circ T$$

so that the spatial perturbations imposed on the source image are invariant to the translation of the source image to the target domain or vice versa. This model is trained adversarially by two components: one maximizes the perturbation while in competition with the network consisting of encodings and decodings, and the other aligns the spatial variation caused by changes, such as object size and background interruptions.

MSPC has some unique features that make it different from other models, for example, CycleGAN and GCGAN, in such a way that it overcomes some of their problems,

especially when the spatial variation between the source and target domains is considerable. Unlike standard constraints, which may be too strong or weak, MSPC focuses on maintaining spatial consistency, thus reducing content distortion and enhancing quality.

From the results, it is clear that one of the major strengths of MSPC is that it is very efficient in dealing with significant domain differences to model spatial transformations. This brings out a lot of translation reality and performs better, refining the base methods than competing state-of-art methods on various benchmarks. That is why it shows much robustness across the datasets and settings, and therefore, the model is very versatile for any different applications. Also worth mentioning is the new benchmark dataset introduced by MSPC, from front face to profile face, presenting real-world challenges in image translation.

However, MSPC comes with its own set of weaknesses. As the model introduces spatial perturbations and adversarial components, the model’s complexity increases, leading to more significant computational requirements and more extensive training. Furthermore, it requires careful optimization and parameter tuning to balance the effects of these spatial perturbations with the translation consistency through fine-tuning the model.

In short, the proposed MSPC includes significant advanced contributions for unpaired image-to-image translation: focusing on spatial perturbation consistency to reduce content distortion and improve translation quality. It considerably adds complexity and computational overhead, but its robustness and effectiveness on numerous benchmarks make this a valuable contribution to the field.

2.2.6 GP-UNIT

This concept was introduced in GP-UNIT (Generative Prior-guided UNsupervised Image-to-image Translation) by Yang et al. (2022)[lit6](#), presented at CVPR. It uses pre-trained class-conditional GANs, such as BigGAN, for guidance within the optimization through generative priors, all to secure better quality and applicability for the final translation.

The GP-UNIT fundamentally borrows the generative priors from pre-trained GANs to capture rich content correspondences between different visual domains. The framework’s functioning can be summed up as going from coarse to fine. Firstly, it distills the generative before capturing robust features of the coarse content that link objects at the abstract semantic level. Then, it learns the fine-level content

features to capture multi-level content correspondences adaptively precisely. This two-stage approach allows GP-UNIT to achieve state-of-the-art effectiveness and quality in translations, even in domains exhibiting significant visual disparities.

In some of the characteristics, GP-UNIT is a deviation from some models since it uses only generative priors. While the CycleGAN model does direct image-level transformations, GP-UNIT builds up on the stage with the available semantic richness of the already pre-trained GANs to form a powerful foundational correspondence between the domains. It will then undergo fine-tuning, adjusting these correspondences specifically for the characteristics of the images in which they are used; this will, in turn, make translations more robust and diverse.

One of the strengths of this model is that it can handle a wide range of image translation tasks with very high fidelity and diversity through generative priors. Source images are translated in an attempt to maintain their semantic integrity while adapting to the style of the target domain. This approach easily enables multimodal translations and also generalizes to classes beyond the pre-trained dataset of the GAN.

The disadvantages of GP-UNIT are also there. It relies on the pre-training of the GANs on the quality and scope of an initial generative model, which can limit its performance if the domain of the pre-trained model is not very diverse. Moreover, the two-stage training process could be pretty resource-intensive and may require a lot of resources and time to get good results.

In conclusion, GP-UNIT is a big step forward for unsupervised image-to-image translation, which utilizes generative priors in a very effective way in both quality and the robustness of translation. Though possibly limited by pre-trained model dependency and computational demands, GP-UNIT still presents itself as a versatile and robust framework for a wide range of high-quality image translations.

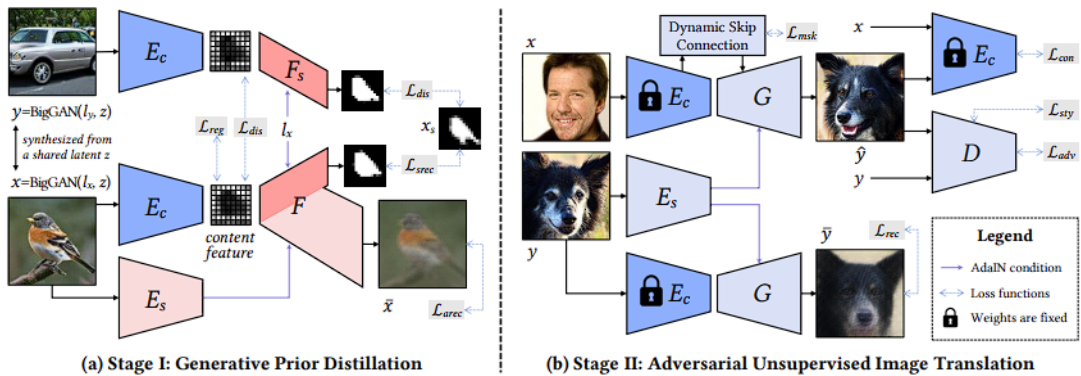


Figure 2.4: GP-UNIT

2.2.7 UNIT

UNIT (Unsupervised Image-to-Image Translation Networks) was introduced in the paper "Unsupervised Image-to-Image Translation Networks" by Li et al. (2017)^{lit8}. This model aims to address the challenge of translating images between domains without paired examples by leveraging a shared latent space.

The basic idea behind UNIT is to assume a partially shared latent space between the source and target domains. Specifically, the model uses Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) to learn a shared latent representation. The framework consists of two VAEs, one for each domain, which share the later layers of the encoder and the initial layers of the decoder. This shared latent space allows for the translation of images from one domain to another by encoding an image from the source domain, decoding it into the shared latent space, and then decoding it back into the target domain.

UNIT differs from other models like CycleGAN by explicitly using the shared latent space assumption, which facilitates the mapping between domains without the need for paired data. This shared latent space ensures that corresponding images in the two domains are mapped to the same latent code, allowing for effective translation even when the domains have significant differences.

The strengths of UNIT include its ability to generate high-quality translations across various tasks, such as street scene image translation, animal image translation, and face image translation. The model also performs well in domain adaptation tasks, achieving state-of-the-art results on several benchmarks. The shared latent space provides a robust framework for translating complex and diverse images between domains.

However, UNIT also has weaknesses. The reliance on a shared latent space can be limiting if the domains are extremely different, making it challenging to find a common representation. The training process is also computationally intensive, requiring careful tuning of the model's hyperparameters to achieve optimal performance. Additionally, the model's complexity can lead to longer training times compared to simpler models like CycleGAN.

In summary, UNIT represents a significant advancement in unsupervised image-to-image translation by leveraging a shared latent space to facilitate translation without paired data. Despite its computational demands and the challenge of finding a shared latent space for very different domains, UNIT's ability to produce high-

quality translations makes it a valuable tool for various image translation and domain adaptation tasks.

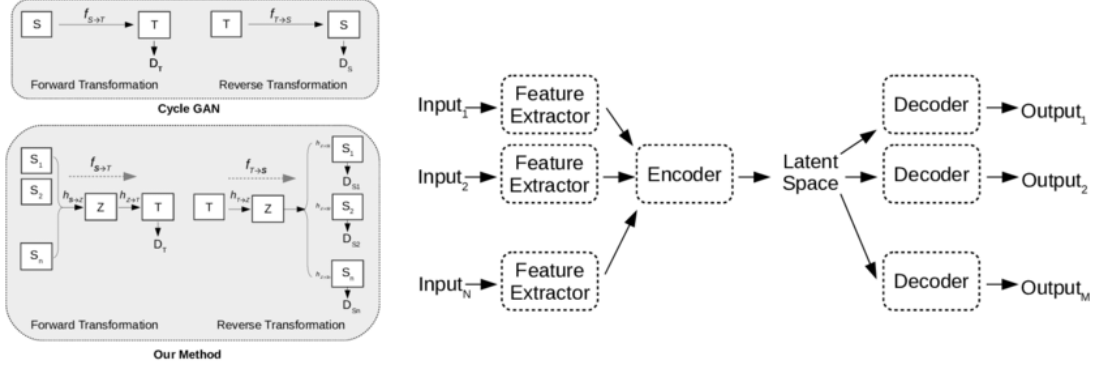


Figure 2.5: UNIT Architecture

2.2.8 MUNIT

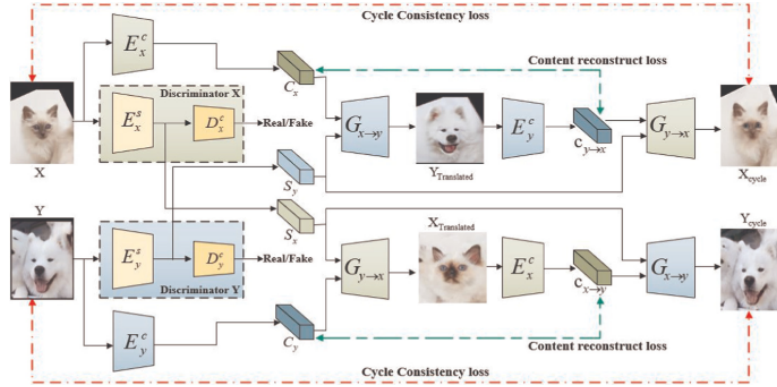


Figure 2.6: MUNIT Architecture

MUNIT (Multimodal Unsupervised Image-to-Image Translation) was introduced in the paper "Multimodal Unsupervised Image-to-Image Translation" by Huang et al. (2018)[lit7](#), presented at ECCV 2018. This model addresses the challenge of translating images between domains without paired examples, while also generating diverse outputs.

The basic idea behind MUNIT is to decompose the image representation into a content code, which is domain-invariant, and a style code, which captures domain-specific properties. The model employs two encoders and two decoders for each domain. During the translation process, the content code of an image from the source domain is combined with a randomly sampled style code from the target domain's style space. This combination is then passed through the decoder of the

target domain to generate the translated image. This approach allows MUNIT to generate multiple plausible outputs for a given input by varying the style code.

MUNIT differs from other models like CycleGAN and UNIT by focusing on multi-modality. While CycleGAN assumes a deterministic one-to-one mapping between domains, MUNIT allows for many-to-many mappings, which is more realistic for complex translation tasks. By explicitly modeling the content and style separately, MUNIT can generate a diverse range of outputs that better capture the variability within each domain.

The strengths of MUNIT include its ability to produce diverse and high-quality image translations. The separation of content and style allows for flexible and controllable translations, enabling users to generate multiple different outputs for the same input. This makes MUNIT particularly useful for applications requiring high variability and creativity, such as artistic style transfer and data augmentation.

However, MUNIT also has weaknesses. The model's complexity, due to the need for multiple encoders and decoders, can lead to increased computational demands and longer training times. Additionally, the quality of the generated images heavily relies on the proper disentanglement of content and style, which can be challenging to achieve and may require careful tuning of the model's parameters.

In summary, MUNIT represents a significant advancement in the field of unpaired image-to-image translation by introducing a multimodal framework that separates content and style. Despite its computational complexity, MUNIT's ability to generate diverse and high-quality translations makes it a valuable tool for various applications in image processing and computer vision.

2.2.9 ITTR

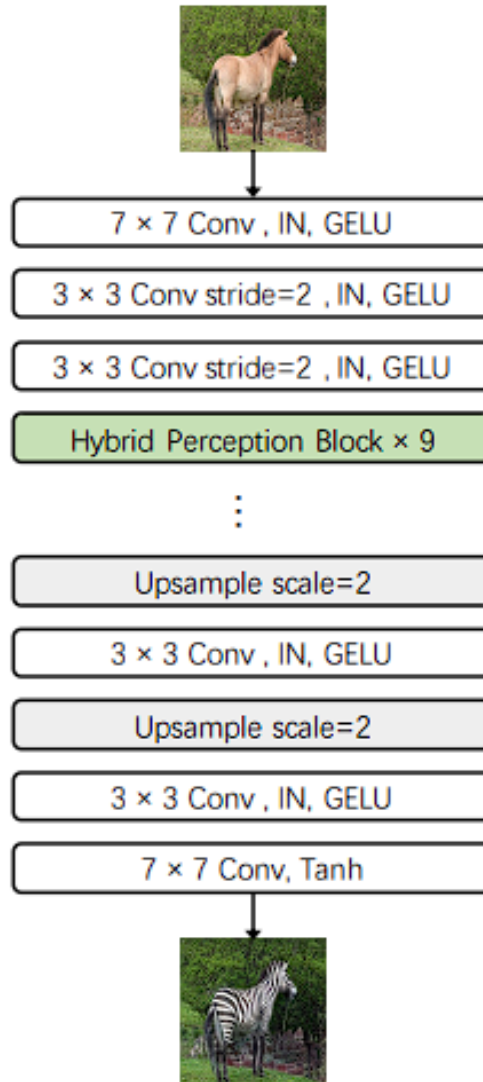


Figure 2.7: ITTR Architecture

ITTR (Image-to-Image Translation with Transformers) was introduced in the paper "ITTR: Unpaired Image-to-Image Translation with Transformers" by Zheng et al. (2022)^{lit9}. The model leverages the capabilities of Vision Transformers (ViTs) to address the limitations of CNN-based generators, particularly in capturing long-range dependencies and global semantics.

The basic idea behind ITTR is to utilize a transformer-based architecture to improve unpaired image-to-image translation. The model consists of two main components:

the Hybrid Perception Block (HPB) and Dual Pruned Self-Attention (DPSA). The HPB captures contextual information from both local and global perspectives by mixing spatial tokens from different receptive fields. The DPSA significantly reduces the computational complexity by pruning less relevant tokens before the self-attention computation, making the model more efficient.

ITTR differentiates itself from other models by incorporating transformers, which are traditionally used in natural language processing and recently adapted for vision tasks. Unlike CNN-based approaches that struggle with long-range dependencies, ITTR’s transformer architecture can effectively capture global context, leading to better image translation quality. This is particularly advantageous in tasks where maintaining consistency across the entire image is crucial.

The strengths of ITTR include its ability to produce high-quality translations by leveraging the global semantic understanding provided by transformers. The hybrid perception block ensures that both local details and global structures are well-represented in the translated images. Additionally, the dual pruned self-attention mechanism optimizes the computational efficiency, allowing ITTR to handle high-resolution images effectively.

However, ITTR also has weaknesses. The introduction of transformer-based components increases the model’s complexity, requiring more computational resources and longer training times. Additionally, fine-tuning the model to achieve optimal performance across different datasets can be challenging due to the intricate architecture and numerous hyperparameters involved.

In summary, ITTR represents a significant advancement in unpaired image-to-image translation by integrating transformers into the generative process. Despite the increased complexity and computational demands, ITTR’s ability to capture long-range dependencies and global context makes it a powerful tool for various image translation tasks.

2.2.10 NICE-GAN

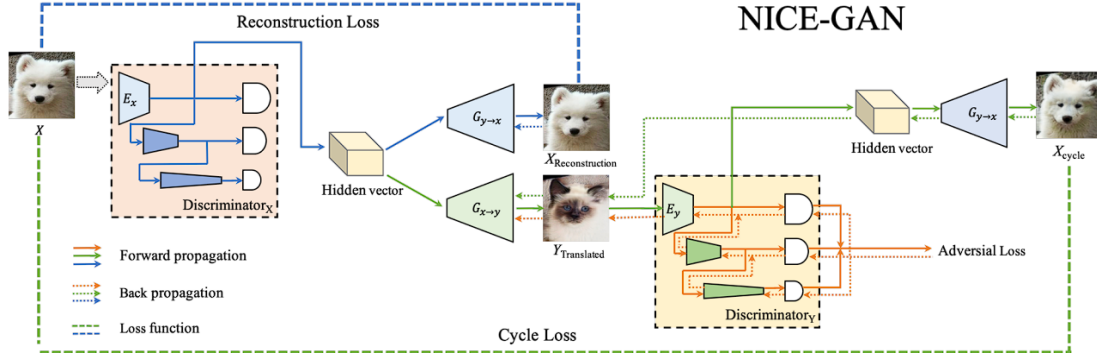


Figure 2.8: NICE-GAN Architecture

NICE-GAN (Noise Injection and Contrastive Estimation GAN) is introduced in the paper "Reusing Discriminators for Encoding: Towards Unsupervised Image-to-Image Translation" by Chen et al. (2020)[3]. This model proposes a novel approach to unsupervised image-to-image translation by reusing the discriminator for encoding, enhancing the efficiency and performance of the translation process.

The basic idea behind NICE-GAN is to decouple the role of the discriminator and the generator in the training process. Unlike traditional GAN frameworks, where the discriminator is discarded after training, NICE-GAN reuses the discriminator for encoding images from the target domain. This approach involves a decoupled training strategy where the encoder is only updated when maximizing the adversarial loss, keeping it frozen otherwise. Additionally, noise injection is used to improve the robustness of the model, and contrastive estimation helps in better aligning the distributions of the source and target domains.

NICE-GAN differentiates itself from other models by introducing the concept of reusing the discriminator for encoding, which reduces the redundancy in the training process and improves the quality of the translations. Traditional GAN models like CycleGAN and UNIT typically use separate networks for encoding and discriminating, which can lead to inefficiencies and increased computational costs. NICE-GAN's approach of coupling these roles enhances the model's efficiency and stability.

The strengths of NICE-GAN include its ability to produce high-quality image translations while being computationally efficient. The reuse of the discriminator for encoding not only simplifies the architecture but also ensures that the features learned by the discriminator are directly used in the translation process, leading to better

alignment between the source and target domains. The noise injection and contrastive estimation further enhance the robustness and accuracy of the model.

However, NICE-GAN also has weaknesses. The reliance on the discriminator for encoding can introduce training instability if not properly managed, requiring careful tuning of the training parameters. Additionally, the decoupled training strategy, while beneficial in some aspects, can make the training process more complex and harder to debug.

In summary, NICE-GAN represents a significant advancement in unsupervised image-to-image translation by reusing the discriminator for encoding and incorporating noise injection and contrastive estimation. Despite the potential challenges in training, its efficient use of resources and improved translation quality make it a promising approach for various applications in image translation and computer vision.

2.2.11 SynDiff

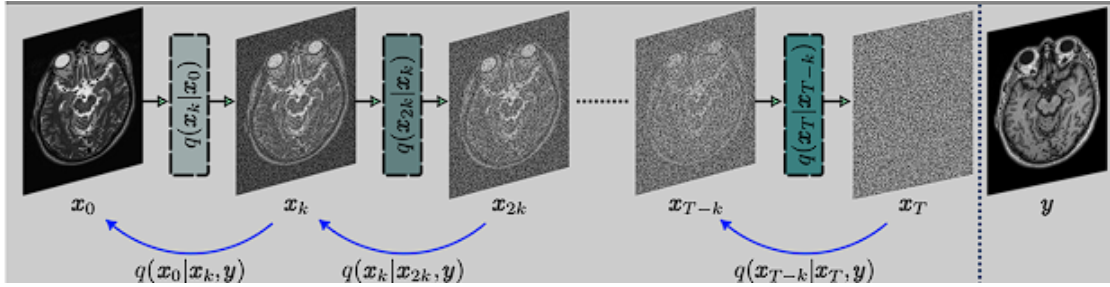


Figure 2.9: SynDiff Architecture

SynDiff (Unsupervised Medical Image Translation with Adversarial Diffusion Models) was introduced in the paper "Unsupervised Medical Image Translation with Adversarial Diffusion Models" by Özbey et al.(2023)[20] This model leverages diffusion models to enhance the performance of medical image translation tasks.

The basic idea behind SynDiff is to combine the strengths of adversarial networks and diffusion models to perform medical image translation without paired training data. The model incorporates a novel adversarial diffusion mechanism that refines image quality through progressive noise reduction and denoising steps. This approach helps in preserving detailed structures and improving the overall visual fidelity of the translated images.

SynDiff differentiates itself from other models by using diffusion processes within an adversarial framework. Traditional GAN-based models often struggle with mode

collapse and instability during training, especially with unpaired data. By integrating diffusion models, SynDiff mitigates these issues, providing more stable and reliable image translation results. The use of adversarial training ensures that the generated images are not only structurally consistent but also visually realistic.

The strengths of SynDiff include its ability to generate high-quality medical images across various tasks such as translating between different MRI modalities (e.g., T1 to T2) and improving the quality of PET images. The diffusion process helps in maintaining fine details and reducing noise, which is crucial in medical imaging. Additionally, SynDiff has been shown to outperform several state-of-the-art methods on multiple benchmark datasets, demonstrating its robustness and versatility.

However, SynDiff also has weaknesses. The integration of diffusion models increases computational complexity and training time, requiring significant computational resources. Fine-tuning the hyperparameters and ensuring the balance between the adversarial and diffusion components can be challenging. Furthermore, while diffusion models enhance stability, they also introduce additional steps that can complicate the training pipeline.

In summary, SynDiff represents a significant advancement in medical image-to-image translation by combining adversarial networks with diffusion models. Despite the increased computational demands, its ability to produce high-quality, realistic medical images makes it a valuable tool for various applications in medical imaging.

2.2.12 Med-GAN

MedGAN (Medical Image Translation using GANs) was introduced in the paper "MedGAN: Medical Image Translation using GANs" by Armanious et al. (2020)^{lit12}. This model focuses on leveraging GANs for various medical image-to-image translation tasks, addressing the limitations of previous approaches that were often application-specific or required non-end-to-end training.

The basic idea behind MedGAN is to create a high-capacity generator architecture, named CasNet, inspired by ResNets, which can be applied to multiple medical imaging tasks without needing application-specific modifications. CasNet chains several fully convolutional encoder-decoder networks with skip connections, progressively refining the input images to produce high-resolution and detailed outputs. MedGAN incorporates a discriminator network that serves as a trainable feature extractor, calculating modified perceptual loss and style-transfer losses to ensure

that the translated images match the target modality in terms of style, texture, and content.

MedGAN differentiates itself from other models by combining the adversarial framework with non-adversarial losses to capture both high and low-frequency components of the desired target modality. This combination enhances the quality and fidelity of the translated medical images. The framework's end-to-end architecture and the inclusion of style-transfer losses make MedGAN versatile and effective for various medical imaging tasks.

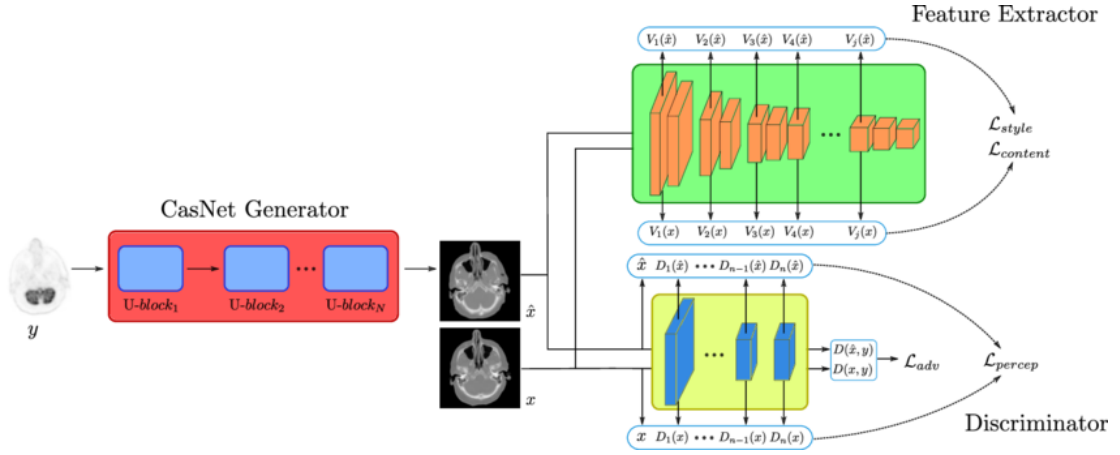


Figure 2.10: Med-Gan Architecture

The strengths of MedGAN include its ability to handle different medical imaging tasks such as PET to CT translation, PET image denoising, and correction of MR motion artifacts without requiring specific modifications to the hyperparameters. The model has demonstrated superior performance in both qualitative and quantitative comparisons with existing translation approaches. Additionally, MedGAN has been evaluated by experienced radiologists, confirming its effectiveness from a medical perspective.

However, MedGAN also has weaknesses. The complexity of the CasNet architecture and the integration of multiple loss functions can lead to increased computational demands and longer training times. The performance of the model also depends on the quality of the pre-trained feature extractors and the proper tuning of hyperparameters, which can be challenging.

In summary, MedGAN represents a significant advancement in medical image-to-image translation by offering a robust and versatile framework that delivers high-quality results across various tasks. Despite its computational complexity, MedGAN's ability to generate realistic and high-fidelity medical images makes it a valuable tool

for the medical imaging community.

2.2.13 SCAM-GAN

SCAM-GAN (Spatial & Channel Attention Mechanism-Generative Adversarial Network) was introduced in the paper "SCAM-GAN: Generating Brain MR Images from CT Scan Data Based on CycleGAN Combined with Attention Module"[17] by Liu et al(2023). This model addresses the challenge of generating MR images from CT scans using unpaired training data, which is particularly useful for patients who cannot undergo MRI due to various constraints.

The basic idea behind SCAM-GAN is to integrate a Spatial and Channel Attention Mechanism (SCAM) with the CycleGAN framework to enhance the quality and consistency of image translations. The model leverages SCAM to focus on important features in both spatial and channel dimensions, improving the generation of detailed and accurate MR images from CT scans.

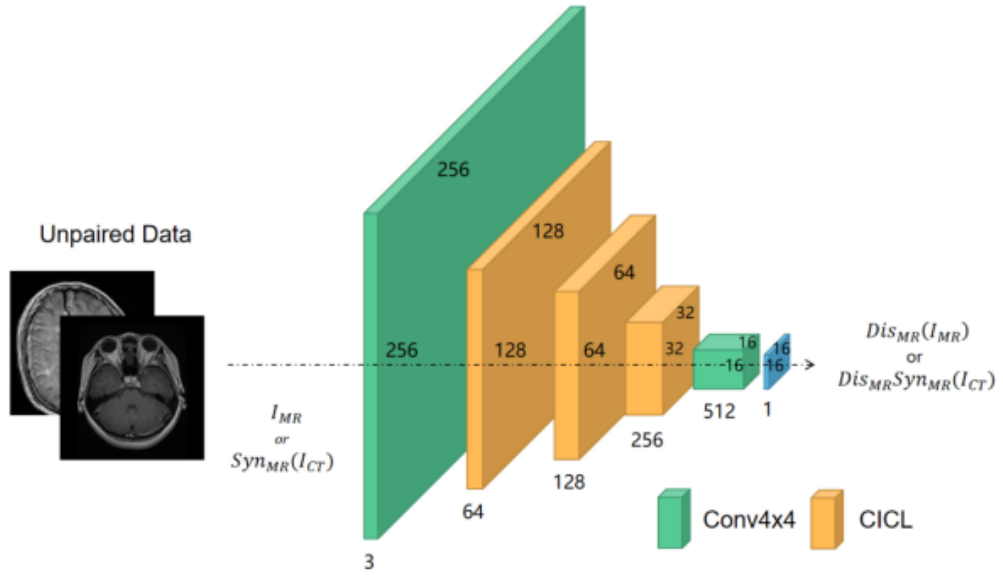


Figure 2.11: SCAM-GAN Discriminator Architecture

SCAM-GAN differentiates itself from other models like CycleGAN by incorporating the attention mechanism, which allows the model to better capture and utilize relevant features in the input images. This results in improved performance compared to using CycleGAN alone, especially in terms of maintaining the anatomical structures and details necessary for medical diagnosis.

The strengths of SCAM-GAN include its ability to generate high-quality MR images that are spatially and anatomically consistent with the original CT scans. The

use of SCAM enhances the model’s attention to critical features, leading to better translation results. SCAM-GAN performs well in both quantitative and qualitative evaluations, demonstrating its effectiveness in medical imaging applications.

However, SCAM-GAN also has weaknesses. The added complexity of the attention mechanism increases the computational requirements and training time. Fine-tuning the model to achieve the best performance across different datasets can be challenging, and the need for substantial unpaired data may limit its applicability in scenarios with limited data availability.

In summary, SCAM-GAN represents a significant advancement in unpaired image-to-image translation by combining CycleGAN with spatial and channel attention mechanisms. Despite its computational demands, SCAM-GAN’s ability to produce high-quality and consistent MR images from CT scans makes it a valuable tool for medical imaging and diagnosis.

2.3 Single Work on Unpaired PET-CT Image Translation

Unpaired PET to CT image translation refers to the process of generating CT images from PET images without the requirement of paired training data, where each PET image corresponds directly to a CT image. This technique is especially important in medical imaging, where acquiring paired datasets can be challenging due to the differences in imaging modalities and the associated costs and risks. For instance, PET (Positron Emission Tomography) provides functional imaging that shows metabolic processes in the body, while CT (Computed Tomography) provides detailed anatomical information.

In clinical practice, multimodal imaging combining PET and CT provides comprehensive information that can enhance diagnosis and treatment planning. However, the scarcity of paired PET/CT data limits the development and application of deep learning models. Unpaired image translation models, such as those based on CycleGAN, address this issue by learning mappings between domains using unpaired datasets. These models can generate synthetic CT images from PET images, facilitating tasks such as data augmentation, improved image analysis, and enhanced diagnostic capabilities.

2.3.1 Model Description

The model for unpaired PET/CT image synthesis of the liver region using CycleGAN is introduced in the paper "*Unpaired PET/CT Image Synthesis of Liver Region using CycleGAN*"[23] by Santini et al(2020). This model aims to generate PET/CT data from 2D slices collected from the liver body region of patients, enhancing the CycleGAN framework with feature matching loss and consistency regularization.

2.3.2 Architecture

The architecture of this modified CycleGAN includes two possible generator network options: UNet and ResNet, with discriminators inspired by the PatchGAN model.

Generator Architecture

- **UNet:** This architecture consists of eight encoding levels, followed by an equal number of up-sampling operations in the decoding part. Instance normalization is used throughout the network. The encoding part employs LeakyReLU activations, while the decoding part uses ReLU activations. Dropout layers are added after the bridge part and after each up-sampling layer to reduce the number of parameters.
- **ResNet:** This architecture comprises nine residual blocks, with each block structured with two 3x3 convolutional layers followed by instance normalization and ReLU activations. The blocks are placed in the middle of a small contracting path and a symmetric up-sampling path.

Discriminator Architecture

The discriminators follow the PatchGAN architecture, producing a tensor where each pixel represents the classification of a single overlapping patch (70x70) in the input image. Instance normalization is replaced by spectral normalization to work effectively with the consistency regularization.

2.3.3 Loss Functions

The model uses a combination of adversarial loss, cycle-consistency loss, feature matching loss, and consistency regularization to improve the stability and quality of the synthesized images.

Adversarial Loss

The adversarial loss for the generators G_X and G_Y is given by:

$$L_{advX}(G_X, D_X) = (1 - D_X(x_i))^2 + (D_X(G_X(y_i)))^2 \quad (2.1)$$

$$L_{advY}(G_Y, D_Y) = (1 - D_Y(y_i))^2 + (D_Y(G_Y(x_i)))^2 \quad (2.2)$$

Cycle-Consistency Loss

This loss ensures that an image translated to the target domain and back to the original domain remains unchanged:

$$L_{CC}(G_X, G_Y, D_X, D_Y) = \|x_i - G_X(G_Y(x_i))\|_1 + \|y_i - G_Y(G_X(y_i))\|_1 \quad (2.3)$$

Feature Matching Loss

The feature matching loss encourages the generators to produce images with features similar to the real images by minimizing the difference between the discriminator feature maps:

$$L_{FM}(G_X, G_Y, D_X, D_Y) = \frac{1}{N-1} \sum_{j=1}^{N-1} \|D_X^j(x) - D_X^j(G_Y(y))\|_2 + \frac{1}{N-1} \sum_{j=1}^{N-1} \|D_Y^j(y) - D_Y^j(G_X(x))\|_2 \quad (2.4)$$

Consistency Regularization

Consistency regularization is applied to the discriminators to maintain their ability to distinguish real from fake images even with augmented data:

$$L_{CR}(D_X, D_Y) = \sum_j \lambda_{CR_j} \|D_X^j(x) - D_X^j(T(x))\|_2 + \sum_j \lambda_{CR_j} \|D_Y^j(y) - D_Y^j(T(y))\|_2 \quad (2.5)$$

The overall loss function for the model with penalty and regularization terms is defined as:

$$L_{PR}(G_X, G_Y, D_X, D_Y) = L_{baseline}(G_X, G_Y, D_X, D_Y) + \lambda_{FM} L_{FM}(G_X, G_Y, D_X, D_Y) + L_{CR}(D_X, D_Y) \quad (2.6)$$

where

$$L_{baseline}(G_X, G_Y, D_X, D_Y) = L_{advX}(G_X, D_X) + L_{advY}(G_Y, D_Y) + \lambda_{CC} L_{CC}(G_X, G_Y, D_X, D_Y) \quad (2.7)$$

λ_{CC} and λ_{FM} are hyperparameters controlling the importance of cycle-consistency and feature matching losses, respectively.

2.4 CycleGAN

CycleGAN, a groundbreaking work by Zhu et al. in 2017[36], introduced an innovative approach utilizing generative adversarial networks (GANs). GANs operate as creative rivals, with one network (the generator) striving to craft realistic images, and the other (the discriminator) discerning between real and generated photos. Taking this rivalry further, CycleGAN employs two generators, each dedicated to a specific domain (e.g., black and white to color). It ensures cycle consistency, maintaining image integrity during translation back and forth between domains, and employs adversarial loss to ensure realistic translations. Figure 2.1 shows cycleGAN's basic architecture

The generator architecture consists of an encoder, transformer, and decoder, with the encoder compressing image features and the transformer utilizing residual blocks for transformations. The discriminator adopts a PatchGAN design, mapping from 256×256 to an NxN array of outputs, distinguishing real or fake patches. The cost function involves adversarial, cycle consistency, and identity losses, ensuring realistic translations and maintaining image characteristics.

The following equations summarize the loss functions:

- **Adversarial Loss:** For LSGAN, the discriminator loss functions are modified to minimize the least squares error rather than the binary cross-entropy. This leads to smoother gradients and better performance.

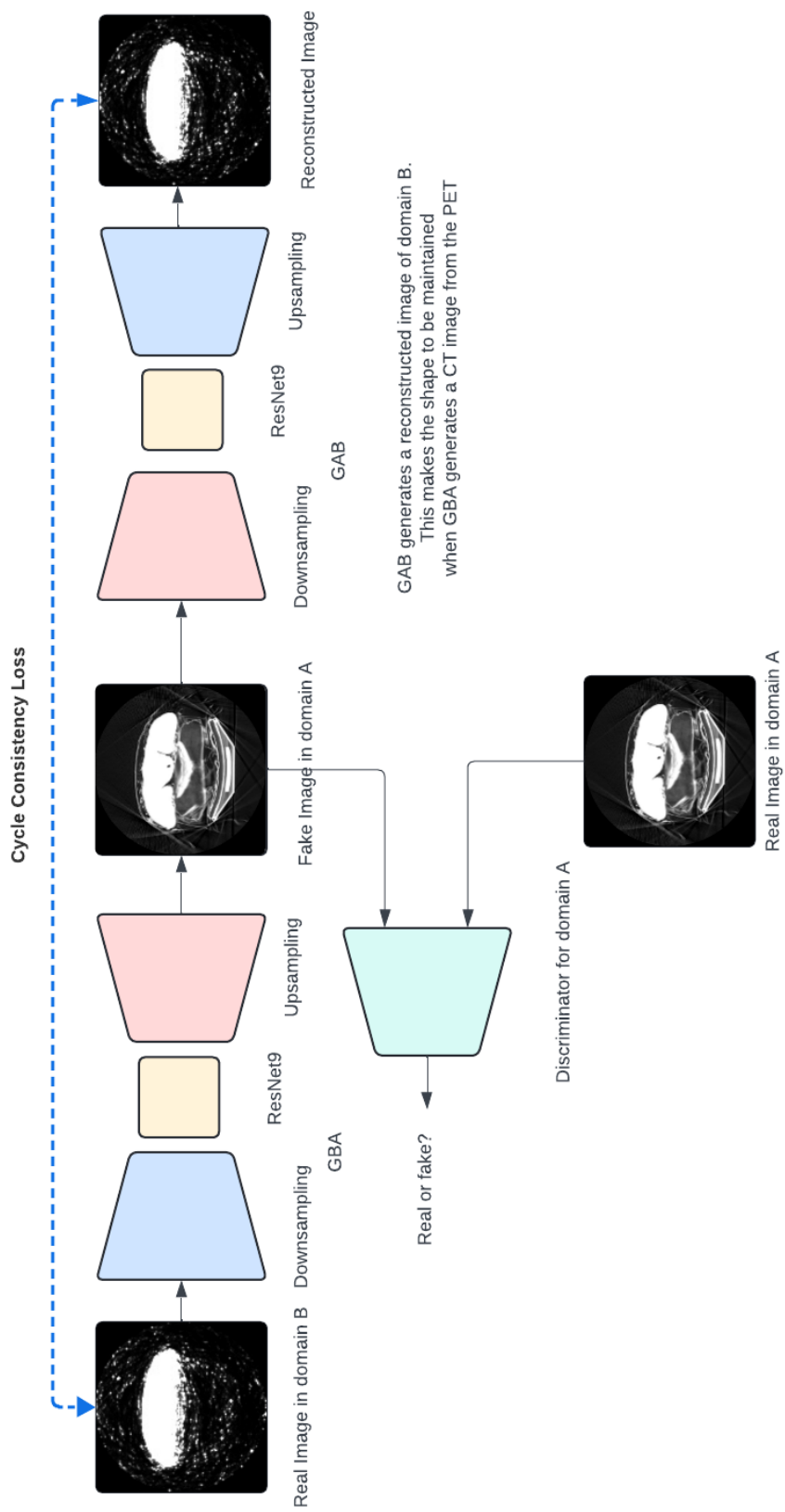
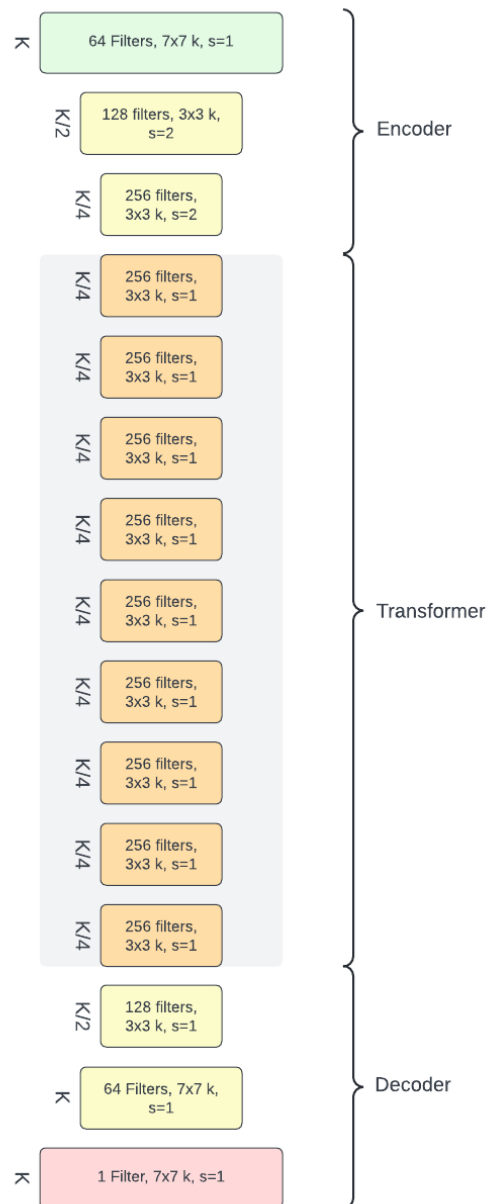
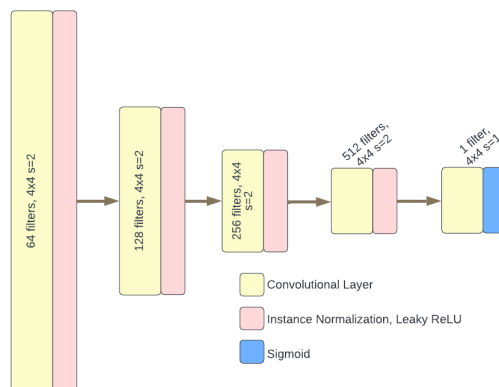


Figure 2.12: CycleGAN Architecture



(a) Generator Architecture



(b) Discriminator Architecture

Figure 2.13: Generator and Discriminator Architecture

- For the discriminator D_Y :

$$\mathcal{L}_{\text{LSGAN}}(D_Y, G, X, Y) = \frac{1}{2} \mathbb{E}_{y \sim p_{\text{data}}(y)} \left[(D_Y(y) - 1)^2 \right] + \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}(x)} \left[D_Y(G(x))^2 \right]$$

- For the discriminator D_X :

$$\mathcal{L}_{\text{LSGAN}}(D_X, F, Y, X) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}(x)} \left[(D_X(x) - 1)^2 \right] + \frac{1}{2} \mathbb{E}_{y \sim p_{\text{data}}(y)} \left[D_X(F(y))^2 \right]$$

The generator loss functions aim to fool the discriminators by making generated images close to the real ones in the least squares sense:

- For the generator G :

$$\mathcal{L}_{\text{LSGAN}}(G, D_Y, X) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}(x)} \left[(D_Y(G(x)) - 1)^2 \right]$$

- For the generator F :

$$\mathcal{L}_{\text{LSGAN}}(F, D_X, Y) = \frac{1}{2} \mathbb{E}_{y \sim p_{\text{data}}(y)} \left[(D_X(F(y)) - 1)^2 \right]$$

- Cycle Consistency Loss: Cycle consistency loss ensures that an image translated to the other domain and back should look like the original image. This loss enforces the cycle consistency:

- For the cycle $X \rightarrow Y \rightarrow X$:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} \left[\|F(G(x)) - x\|_1 \right]$$

- For the cycle $Y \rightarrow X \rightarrow Y$:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{y \sim p_{\text{data}}(y)} \left[\|G(F(y)) - y\|_1 \right]$$

Combining these, the total cycle consistency loss is:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} \left[\|F(G(x)) - x\|_1 \right] + \mathbb{E}_{y \sim p_{\text{data}}(y)} \left[\|G(F(y)) - y\|_1 \right]$$

- Identity Loss: Identity loss helps to preserve the color composition between the input and output. This is particularly useful when the input and output

images are expected to be similar. The identity loss is:

- For generator G :

$$\mathcal{L}_{\text{identity}}(G, Y) = \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(y) - y\|_1]$$

- For generator F :

$$\mathcal{L}_{\text{identity}}(F, X) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(x) - x\|_1]$$

CycleGAN has its merits. It is very flexible, and the translation of images from one domain to another is usually very effective, usually without the need for paired data. This is particularly useful for tasks such as style transfer, whereby images can be rendered into another domain of style, provided that the content remains. Furthermore, CycleGAN has been applied to problems such as artistic style transfer, photo enhancement, and object transfiguration.

Some of the weaknesses of this model are that it may become unstable in training and can generate artifacts in translated images. As it treats all the pixels the same way it is very weak to properly establish a relationship between the two domains efficiently. Also its generator architecture is specially tuned for style transfer and susceptible to big structural difference. Another con is that the quality of the pictures of the output maybe not so good, and there may be no details in some areas present in the original images. All in all, instability issues in training and the problem of image quality in CycleGAN open room for novel insights in methods for image translation due to unpaired training. Its ingenious cycle consistency serves as a base for further research and development in GAN-based image translation models.

2.5 Contrastive Learning for Unpaired Image-to-Image Translation

Contrastive Learning for Unpaired Image-to-Image Translation (CUT) is introduced in the paper "*Contrastive Learning for Unpaired Image-to-Image Translation*" [21] by Park et al(2020). This model introduces a contrastive learning approach to improve the quality and diversity of unpaired image-to-image translation tasks.

In this approach, a contrastive loss is used to guide the learning procedure so that the generated images and their inputs are made similar in a feature space. The following

approach maximizes mutual information between input and corresponding output images translated from them, preserving essential content while allowing stylistic transformations.

Most distinctive from methods like CycleGAN is that Contrastive Learning for Unpaired Image-to-Image Translation aligns features of different domains directly by contrast to CycleGAN. While CycleGAN uses cycle-consistency loss to ensure the translation can be reversed accurately, this model uses contrastive loss for directly learning feature correspondences, which can, in return, yield more stable and coherent translations.

Enables the production of high-quality translations by the model through the inclination of the significant features of the input images, allowing for significant stylistic changes. However, the incorporation of contrastive learning further complicates the whole training process, making it heavy in terms of computation. The contrastive loss parameters need proper fine-tuning to get the best performance, which is difficult.

2.5.1 Architecture

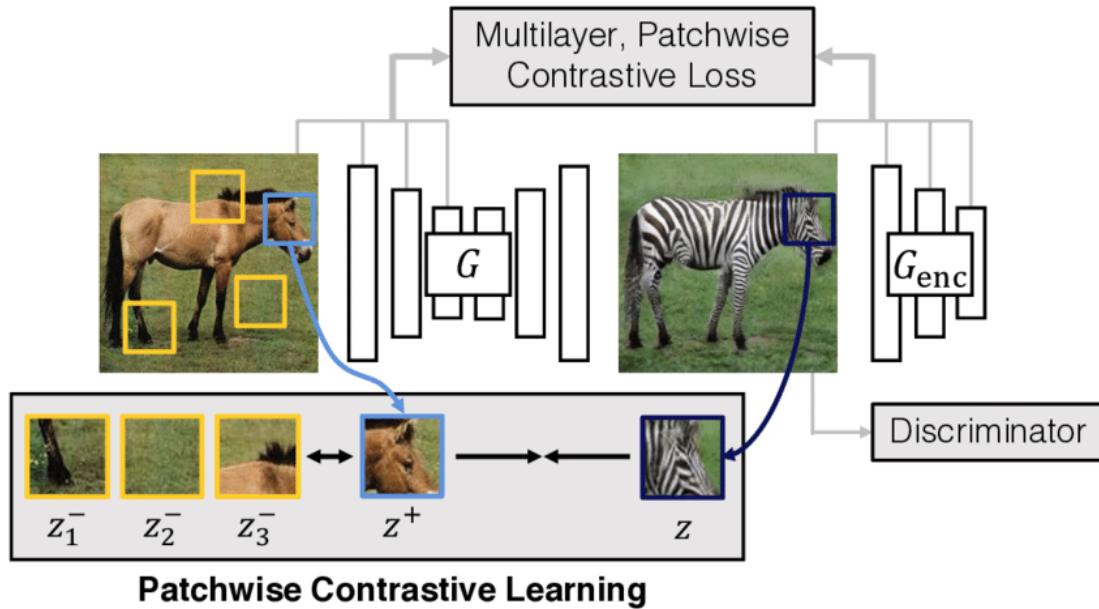


Figure 2.14: CUT architecture

The CUT architecture combines the strengths of GANs with contrastive learning to enhance feature representation and achieve high-quality image translations. The architecture includes a generator and a discriminator, similar to traditional GANs, but with the addition of a PatchNCE loss to enforce better feature matching.

Generator Architecture

- The generator G consists of several convolutional layers to extract features from the input image.
- Residual blocks are used to maintain the high-level features and improve image quality.
- The generator uses transposed convolutions for upsampling and reconstructing the image in the target domain.
- Instance normalization and ReLU activations are applied throughout the network.
- The output layer uses Tanh activation to produce the final translated image.

Discriminator Architecture

- The discriminator D is a convolutional neural network that distinguishes real images from generated images.
- It consists of multiple convolutional layers with stride and padding to down-sample the input image.
- LeakyReLU activations are used for non-linearity.
- The output layer uses sigmoid activation to produce a probability score indicating the likelihood of the image being real.

2.5.2 Loss Functions

CUT uses a combination of adversarial loss, contrastive loss (PatchNCE loss), and identity loss to ensure high-quality translations and consistency in feature representations.

Adversarial Loss

The adversarial loss encourages the generator to produce realistic images that can fool the discriminator.

$$L_{GAN}(G, D, X, Y) = \mathbb{E}_{y \sim p_{data}(y)}[\log D(y)] + \mathbb{E}_{x \sim p_{data}(x)}[\log(1 - D(G(x)))] \quad (2.8)$$

Contrastive Loss (PatchNCE Loss)

The PatchNCE loss enforces consistency in the features extracted by the generator from the source and target domains. It uses a contrastive learning approach to maximize the similarity between corresponding patches in the translated and original images while minimizing the similarity between non-corresponding patches.

$$L_{PatchNCE}(G, X, \hat{X}) = -\mathbb{E}_{x \sim p_{data}(x), \hat{x} \sim G(x)} \left[\log \frac{\exp(\text{sim}(z_x, z_{\hat{x}})/\tau)}{\sum_{z' \in \text{patches}} \exp(\text{sim}(z_x, z')/\tau)} \right] \quad (2.9)$$

Here, z_x and $z_{\hat{x}}$ are the features of the original and translated patches, respectively, sim denotes the similarity function (e.g., cosine similarity), and τ is a temperature hyperparameter.

Identity Loss

The identity loss ensures that the generator preserves the identity of images that are already in the target domain.

$$L_{identity}(G, Y) = \mathbb{E}_{y \sim p_{data}(y)} [\|G(y) - y\|_1] \quad (2.10)$$

The total loss function for CUT is a weighted sum of the adversarial, PatchNCE, and identity losses:

$$L(G, D) = L_{GAN}(G, D, X, Y) + \lambda_{PatchNCE} L_{PatchNCE}(G, X, \hat{X}) + \lambda_{identity} L_{identity}(G, Y) \quad (2.11)$$

Here, $\lambda_{PatchNCE}$ and $\lambda_{identity}$ are hyperparameters that control the importance of the PatchNCE and identity losses relative to the adversarial loss.

2.6 Different Generator Architectures

2.6.1 ResNet-9

ResNet-9 (Residual Network with 9 layers) is introduced in the context of various deep learning applications, especially in computer vision tasks. The original ResNet architecture, including ResNet-9, is detailed in the paper "Deep Residual Learning

for Image Recognition" by Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun [8]. ResNet-9 is a simplified version of the deeper ResNet architectures designed to reduce computational complexity while maintaining performance.

The basic idea of ResNet-9 involves using residual blocks that allow the network to learn residual functions with reference to the layer inputs. In a ResNet-9 architecture, there are nine layers, including both convolutional layers and residual blocks. The residual blocks contain shortcut connections that enable the direct addition of the input to the output of a series of convolutional layers, facilitating the training of deeper networks by addressing the vanishing gradient problem.

ResNet-9 differs from other architectures by its use of a smaller number of layers while still incorporating the key concept of residual connections. This makes it less complex and computationally demanding compared to deeper versions of ResNet, such as ResNet-50 or ResNet-101. The reduced number of layers makes ResNet-9 particularly suitable for tasks requiring lower computational resources, while still benefiting from the advantages of residual learning.

The strengths of ResNet-9 include its simplicity and efficiency, making it easier to implement and faster to train than deeper networks. It is effective for smaller-scale applications and can serve as a strong baseline model. However, the weaknesses of ResNet-9 include its limited capacity to model very complex functions compared to deeper architectures. For very large and intricate datasets, deeper networks might be necessary to capture more detailed patterns and achieve higher performance.

Overall, ResNet-9 is a valuable architecture for scenarios where computational efficiency is critical, providing a balance between performance and complexity.

2.6.2 ResNet-18

ResNet-18 (Residual Network with 18 layers) is introduced in the paper "Deep Residual Learning for Image Recognition" [8] by Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. ResNet-18 is a part of the ResNet family of architectures designed to address the vanishing gradient problem, allowing for the training of much deeper networks by introducing residual connections.

The basic idea of ResNet-18 involves the use of residual blocks that allow the network to learn residual functions with reference to the layer inputs. The architecture includes 18 layers, composed of convolutional layers, batch normalization, and ReLU activation functions, organized into several residual blocks. Each residual

block includes identity mappings where the input is passed directly to the output, facilitating the training process by ensuring gradients can flow through the network without diminishing.

ResNet-18 differs from other architectures by incorporating these residual connections, which allow for the training of deeper networks without suffering from the degradation problem seen in traditional deep networks. Unlike very deep versions like ResNet-50 or ResNet-101, ResNet-18 strikes a balance between depth and computational efficiency, making it suitable for tasks that require deeper networks than ResNet-9 but not as computationally intensive as larger ResNets.

The strengths of ResNet-18 include its ability to train deeper networks without the vanishing gradient problem, leading to better performance on complex tasks compared to shallower networks. It is computationally more efficient than larger ResNet variants while still providing significant improvements in accuracy and performance for various image recognition tasks.

However, the weaknesses of ResNet-18 include its limited ability to capture extremely complex patterns compared to much deeper networks. For highly intricate tasks that require capturing fine-grained details and very large datasets, deeper architectures like ResNet-50 or ResNet-101 might be necessary to achieve optimal performance. Additionally, while more efficient than its deeper counterparts, ResNet-18 still requires more computational resources than very shallow networks like ResNet-9.

Overall, ResNet-18 provides a balanced approach, offering a robust architecture suitable for many practical applications where deeper learning is beneficial, but extremely deep networks are not feasible due to resource constraints.

2.6.3 UNET-128

UNET-128 is introduced in the paper "U-Net: Convolutional Networks for Biomedical Image Segmentation" by Olaf Ronneberger, Philipp Fischer, and Thomas Brox [22]. While the original U-Net was designed for biomedical image segmentation, UNET-128 is a variant used primarily in the context of GANs, particularly for image-to-image translation tasks, such as in the Pix2Pix framework.

The basic idea of UNET-128 involves an encoder-decoder structure with skip connections between corresponding layers in the encoder and decoder. The encoder gradually reduces the spatial dimensions through convolutional layers and pooling,

capturing context and features. The decoder then upsamples the features back to the original resolution. The skip connections allow the network to leverage both high-level and low-level features, improving the quality of the output images.

UNET-128 differs from other architectures by specifically tailoring the U-Net framework to 128x128 image resolution tasks. The original U-Net was designed for medical image segmentation with different input sizes. UNET-128's architecture is optimized for handling images at this specific resolution, making it particularly suitable for applications requiring moderate image resolutions.

The strengths of UNET-128 include its ability to preserve fine details in the generated images due to the skip connections, which combine high-level abstract features with low-level detailed features. This makes UNET-128 highly effective for tasks requiring precise reconstruction, such as segmentation and image-to-image translation. Additionally, the relatively shallow depth compared to larger U-Net variants means it requires less computational power, making it suitable for scenarios with limited resources.

However, the weaknesses of UNET-128 include its limited ability to capture very complex features and patterns compared to deeper networks. For high-resolution images or tasks requiring more detailed feature extraction, larger variants of U-Net or different architectures might be more effective. Moreover, the model's performance heavily depends on the quality and quantity of the training data, requiring well-annotated datasets to achieve optimal results.

Overall, UNET-128 is a robust and efficient architecture for tasks involving moderate-resolution image generation and translation, offering a good balance between performance and computational efficiency.

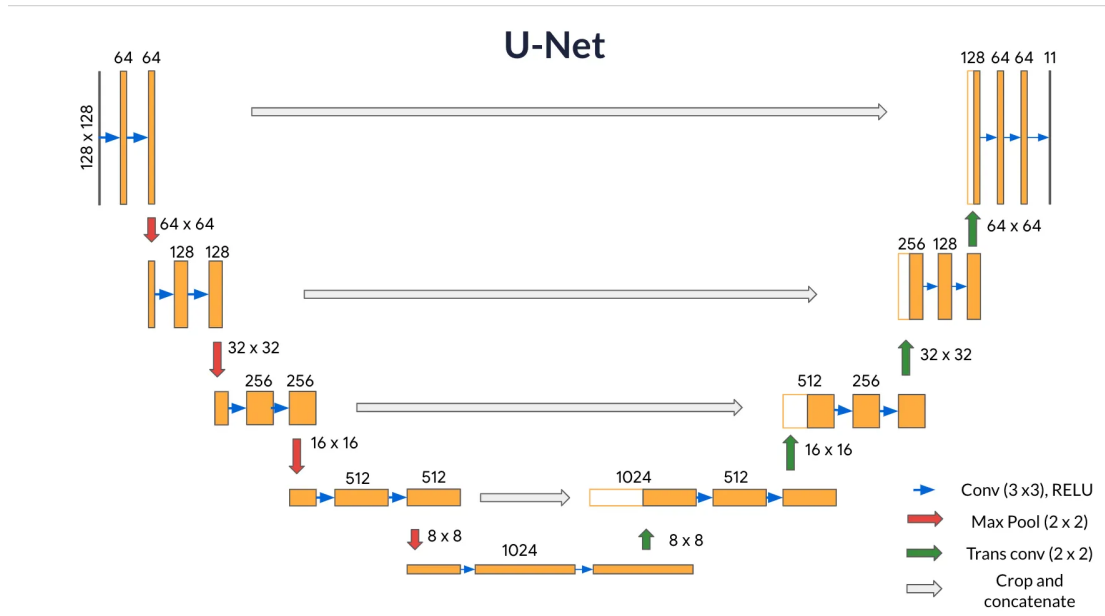


Figure 2.15: Unet-128 Architecture

2.6.4 UNET-256

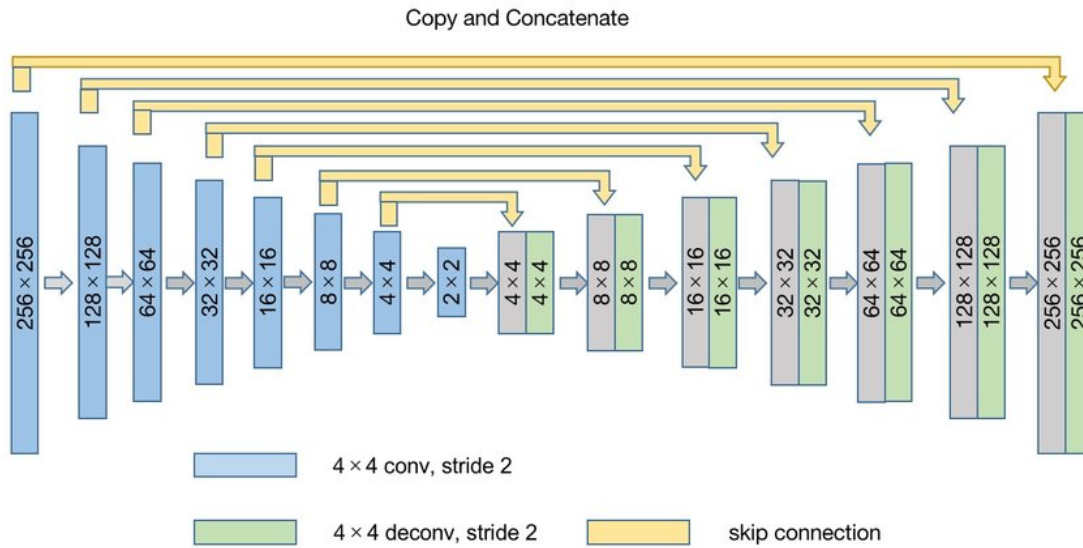


Figure 2.16: UNET-256 Architecture

UNET-256 is introduced in the context of various deep learning applications, particularly for image-to-image translation tasks, extending from the original U-Net architecture described in the paper "U-Net: Convolutional Networks for Biomedical Image Segmentation" by Olaf Ronneberger, Philipp Fischer, and Thomas Brox [22]. UNET-256 is an adaptation of the U-Net architecture, designed to handle higher resolution images, specifically 256x256 pixels.

The basic idea of UNET-256 involves an encoder-decoder structure with skip connections between corresponding layers in the encoder and decoder. The encoder compresses the input image into a lower-dimensional representation through a series of convolutional and pooling layers, while the decoder upsamples this representation back to the original resolution using convolutional and upsampling layers. The skip connections directly transfer spatial information from the encoder to the decoder, enabling the network to generate high-resolution output images that maintain fine details.

UNET-256 differs from other architectures by optimizing the U-Net framework for handling 256x256 image resolutions. This variant is particularly useful in scenarios requiring detailed image translation or reconstruction, such as in the Pix2Pix framework. Unlike simpler models, UNET-256 can manage higher resolution inputs and outputs, making it suitable for more complex image generation tasks.

The strengths of UNET-256 include its ability to preserve fine details in the generated images due to the skip connections, which allow for the integration of both high-level and low-level features. This makes UNET-256 highly effective for tasks requiring precise image reconstruction and segmentation. Additionally, the relatively deep architecture allows it to capture more complex features compared to smaller U-Net variants.

However, the weaknesses of UNET-256 include its increased computational complexity and memory usage compared to smaller models like UNET-128. Training UNET-256 requires more computational resources and can be more time-consuming. Furthermore, the model's performance is highly dependent on the availability of large, well-annotated datasets, as it requires substantial data to learn the intricate patterns necessary for high-quality image generation.

Overall, UNET-256 provides a powerful and flexible architecture for high-resolution image-to-image translation tasks, offering a good balance between detail preservation and computational efficiency. It is well-suited for applications where detailed image generation is crucial, though it requires significant computational resources and data to achieve optimal performance.

2.6.5 Vision Transformer

The Vision Transformer (ViT) is introduced in the paper "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale" by Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai [4], Thomas

Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. ViT brings the transformer architecture, originally designed for natural language processing (NLP), to the realm of computer vision, offering a new approach to image recognition tasks.

The basic idea of ViT involves dividing an image into a sequence of fixed-size patches, typically 16x16 pixels, and then embedding each patch into a lower-dimensional space. These embeddings are treated as tokens, similar to words in NLP, and are fed into a standard transformer encoder. The transformer encoder processes these tokens through self-attention mechanisms and feed-forward layers, capturing the global context and dependencies between patches. Finally, a classification token is added to the sequence, and its representation is used for the final classification task.

ViT differs from other architectures by relying entirely on the transformer architecture without using convolutional layers, which are a staple in traditional convolutional neural networks (CNNs). This approach allows ViT to capture long-range dependencies more effectively than CNNs, which typically rely on local receptive fields. Additionally, the use of transformers enables ViT to leverage the benefits of pre-training on large-scale datasets, followed by fine-tuning on specific tasks.

The strengths of ViT include its ability to model global relationships between different parts of an image, leading to superior performance on large-scale image recognition tasks when pre-trained on extensive datasets. ViT has demonstrated state-of-the-art results on various benchmarks, showing that transformers can outperform traditional CNNs in certain scenarios. Moreover, ViT's architecture is highly scalable, benefiting significantly from increased data and computational resources.

However, the weaknesses of ViT include its reliance on very large datasets and substantial computational resources for pre-training. Without extensive pre-training, ViT may not perform as well as CNNs on smaller datasets. Additionally, the quadratic complexity of the self-attention mechanism with respect to the input size can lead to higher computational costs compared to convolutional architectures, especially for high-resolution images.

Overall, the Vision Transformer represents a significant advancement in image recognition, leveraging the strengths of transformers to model global image context effectively. It offers a promising alternative to traditional CNNs, particularly for tasks with abundant data and computational resources.

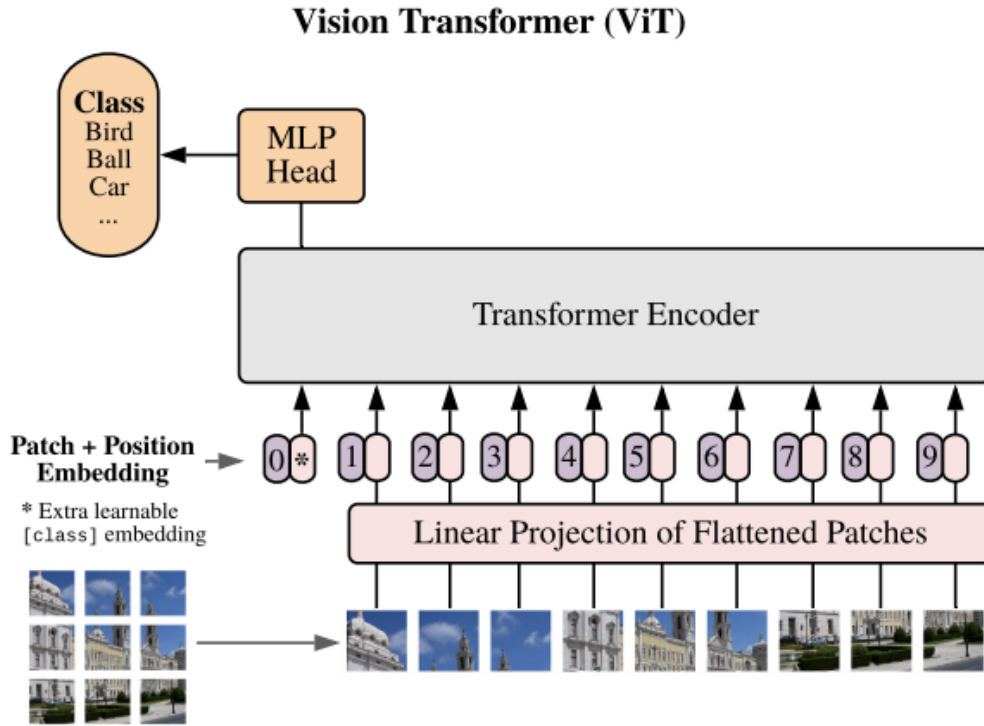


Figure 2.17: Vision Transformer Architecture

2.7 Attention Mechanism

Attention mechanisms, inspired by the human visual attention system, have undergone a transformative journey in reshaping various computer vision tasks, including image classification, object detection, and image generation. This comprehensive review delves into the evolution of attention mechanisms, their influence on different tasks, and potential avenues for future research. Figure 2.2 shows basic attention mechanism working.

The early applications primarily concentrated on integrating attention into recurrent neural networks for tasks such as video and image captioning. Xu et al. (2015) introduced the "Show and Tell" model [25], combining convolutional neural networks (CNNs) with recurrent neural networks and attention, achieving notable success in image captioning. The rise of self-attention mechanisms, as exemplified by Vaswani et al.'s (2017) "Attention is All You Need" [24] with the Transformer architecture, marked a significant shift. Dosovitskiy et al. (2017) [4] adapted this concept to computer vision, showcasing its efficacy in image classification and object detection.

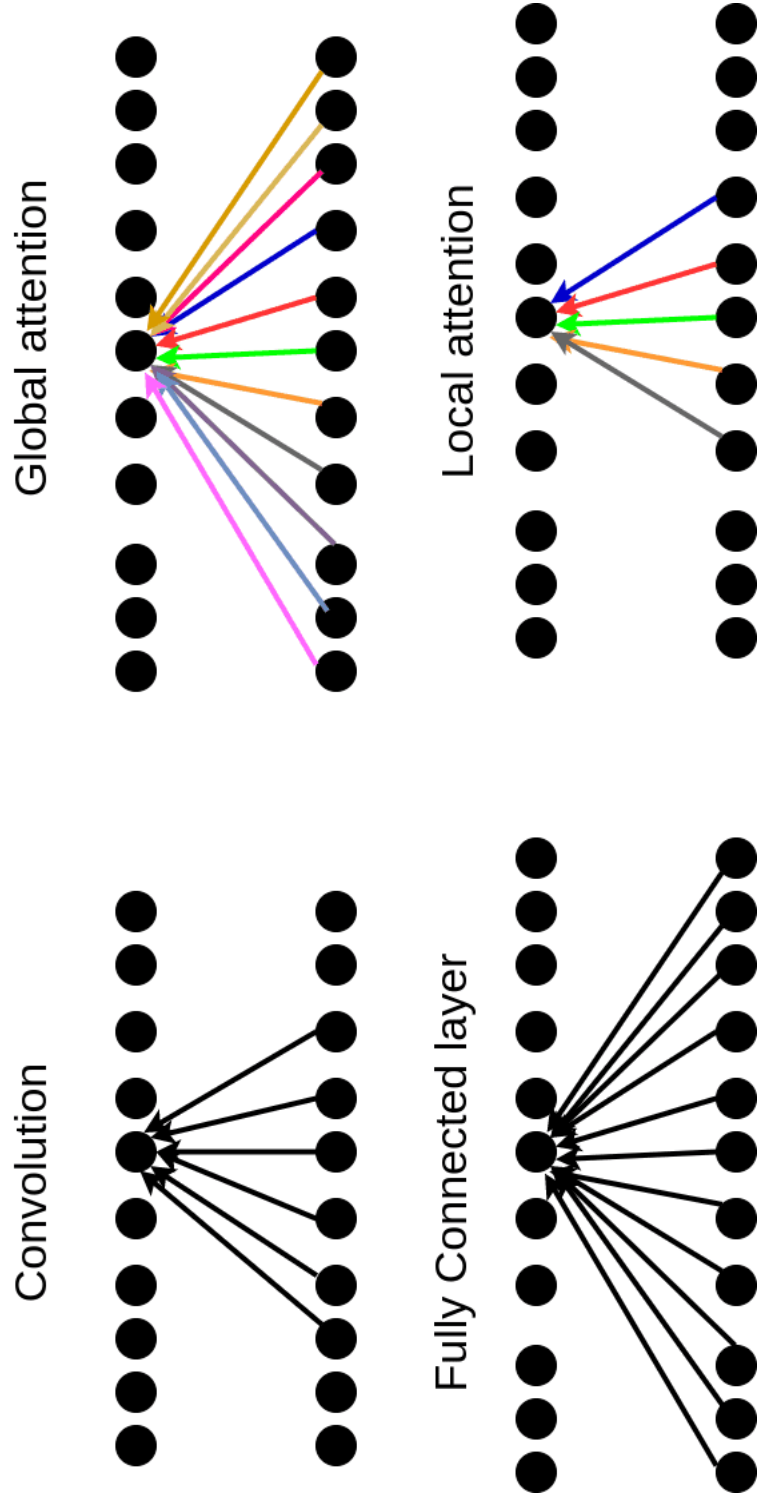


Figure 2.18: Attention Mechanism

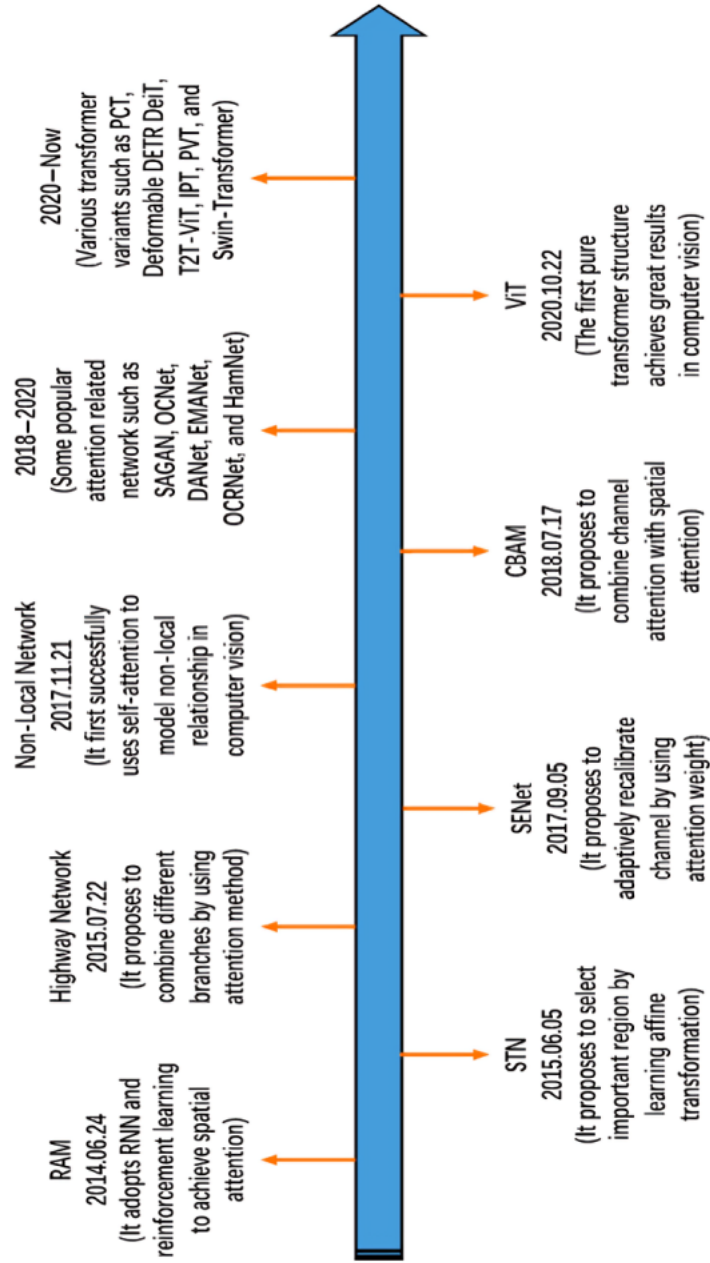


Figure 2.19: Different Attention Mechanisms in Computer Vision

Beyond self-attention, diverse attention types have emerged, each catering to specific needs. Channel attention focuses on relevant feature channels, improving detail retention and texture synthesis (e.g., SENet [9]), while spatial attention selects crucial spatial regions, enhancing saliency and object detection (e.g., Atrous Spatial Pyramid Pooling [35]). Temporal attention captures dependencies between frames in videos, enhancing video understanding and action recognition (e.g., Non-local Neural Networks [27]).

The impact of attention mechanisms on various applications is profound. In image classification, ResNet with attention-based modules, exemplified by ResNext [29], achieved state-of-the-art results. Object detection saw improvements with attention mechanisms like Deformable DETR [38], enhancing accuracy and pose estimation. For image generation, models like AttnGAN [30] and SAGAN[33] incorporated attention for finer detail generation and improved coherence. Figure 2.3 shows different attention mechanisms used in computer visions by timeline[7]

However, challenges persist, including computational cost in large-scale deployments, the need for interpretability to understand decision-making processes, and the task-specific adaptation of attention modules. Addressing these challenges, along with exploring attention fusion possibilities, combining different attention types, promises richer feature representations and better results.

In conclusion, attention mechanisms have become integral to computer vision, significantly elevating performance and unlocking new research possibilities. As attention mechanisms continue to evolve and adapt, addressing computational efficiency, enhancing interpretability, and tailoring attention to specific tasks will be crucial for future advancements, paving the way for groundbreaking applications in computer vision and beyond.

2.8 Attention Mechanism in GAN

The rise of attention mechanisms has revolutionized various domains, including machine translation and computer vision. Their ability to capture long-range dependencies and model relationships between distant elements has proven particularly valuable in the realm of Generative Adversarial Networks (GANs).

The initial attempts to incorporate attention into GANs primarily focused on the discriminator side. Papers like "Spectral Normalization for Generative Adversarial Networks" by Miyato et al. (2018) [19] and "Progressive Growing of GANs for

Improved Quality, Stability, and Variation"[13] by Karras et al. (2017) employed self-attention modules within the discriminator to better analyze global image coherence and prevent artifacts.

The application of attention in CycleGAN gained traction with the work of Xu et al. in "AttnGAN: Attentive Generative Adversarial Networks" (2018)[30]. Their model incorporated channel-wise attention in both generators, allowing features to selectively focus on relevant information from other channels, leading to improved detail retention and texture synthesis. Similarly a global self-attention mechanism can be used in the generator to capture long-range dependencies and enhance image consistency across domains.

Recent advances explore attention mechanisms beyond solely self-attention. For instance, [31] leverages a hybrid encoder-decoder architecture with separate global and local attention modules for image inpainting tasks. This framework captures contextual information across the entire image while simultaneously attending to local details, resulting in high-quality reconstructions. Also SCAM-GAN[17] introduces spatial & channel attention mechanism in the discriminator network which follows CBAM (Convolutional Block Attention Module)[28] which has superior accuracy in MR to CT image synthesis than other paired methods & cycleGAN.

The incorporation of attention mechanisms in image generation processes yields several significant benefits. Firstly, attention facilitates the improvement of image quality by directing the generator's focus towards relevant features and relationships within the input data. This results in generated images with sharper details, improved texture synthesis, and enhanced overall coherence. Additionally, attention mechanisms excel at modeling long-range dependencies in images, allowing generators to capture intricate relationships between distant regions. This capability surpasses the limitations of traditional convolutional layers, thereby contributing to better global consistency in the generated images. Furthermore, attention plays a pivotal role in achieving controllable image generation. By integrating additional information, such as conditioning vectors or class labels, attention allows for a guided generation process, enabling greater control over the characteristics and attributes of the generated output. In essence, attention mechanisms prove instrumental in elevating the quality, spatial coherence, and controllability of generated images.

Despite their impressive advantages, attention mechanisms in CycleGAN and GANs aren't without challenges. The computational cost can soar for high-resolution im-

ages, demanding exploration of efficient implementations and alternative architectures. Deciphering how these models make decisions and which features they favor remains a puzzle, making visualization and interpretation techniques crucial for improved debugging and control. Finally, attention mechanisms designed for one task might not shine in another, so developing context-aware modules tailored to specific applications holds the key to further performance gains.

In essence, while attention mechanisms have unlocked exciting possibilities, mastering their computational efficiency, unraveling their decision-making process, and tailoring them to specific tasks are crucial steps on the path to even more impressive image generation advancements.

2.9 Additional Loss Functions

2.9.1 SSIM Loss

Overview

The Structural Similarity Index Measure (SSIM) is a perceptual metric used to measure the similarity between two images. Unlike traditional metrics such as Mean Squared Error (MSE), which quantify absolute differences, SSIM considers changes in structural information, luminance, and contrast. This makes SSIM more aligned with human visual perception for image quality assessment.

SSIM Calculation

The SSIM index between two image patches $\mathbf{x}$ and $\mathbf{y}$, each of size $N \times N$, is defined as:

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_{\mathbf{x}}\mu_{\mathbf{y}} + C_1)(2\sigma_{\mathbf{xy}} + C_2)}{(\mu_{\mathbf{x}}^2 + \mu_{\mathbf{y}}^2 + C_1)(\sigma_{\mathbf{x}}^2 + \sigma_{\mathbf{y}}^2 + C_2)},$$

where:

$$\begin{aligned}\mu_x &= \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N x_{ij}, & \mu_y &= \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N y_{ij}, \\ \sigma_x^2 &= \frac{1}{N^2 - 1} \sum_{i=1}^N \sum_{j=1}^N (x_{ij} - \mu_x)^2, & \sigma_y^2 &= \frac{1}{N^2 - 1} \sum_{i=1}^N \sum_{j=1}^N (y_{ij} - \mu_y)^2, \\ \sigma_{xy} &= \frac{1}{N^2 - 1} \sum_{i=1}^N \sum_{j=1}^N (x_{ij} - \mu_x)(y_{ij} - \mu_y),\end{aligned}$$

and $C_1 = (k_1 L)^2$ and $C_2 = (k_2 L)^2$ are constants to stabilize the division with weak denominators, with L being the dynamic range of the pixel values (e.g., 255 for 8-bit images), and $k_1 = 0.01$ and $k_2 = 0.03$ by default.

SSIM Loss

The SSIM loss function is utilized in the context of optimizing machine learning models, particularly in image reconstruction and enhancement tasks. It is defined as:

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\mathbf{x}, \mathbf{y}),$$

where $\mathbf{x}$ represents the ground truth image patch and $\mathbf{y}$ denotes the predicted image patch. This loss function aims to maximize the SSIM index, thereby ensuring that the predicted image maintains structural similarity to the ground truth image.

Usage in Image to Image translation

The Structural Similarity Index (SSIM) loss function has been widely used in image-to-image translation and other image processing tasks to enhance the perceptual quality and structural fidelity of generated images. In the study "Loss Functions for Image Restoration with Neural Networks," [34] SSIM loss was employed for tasks such as denoising and demosaicking. The research highlighted that SSIM, being a perceptually motivated error function, significantly improves the visual quality of images by focusing on structural details rather than just pixel-wise differences. Additionally, the paper "HistoHDR-Net: Histogram Equalization for Single LDR to HDR Image Translation" [2] used SSIM loss to measure the structural similarity between LDR and HDR images. This incorporation of SSIM loss helped the

model achieve better performance in terms of maintaining structural consistency and enhancing perceptual quality, as demonstrated by higher SSIM scores compared to other methods. Furthermore, in the context of super-resolution tasks using Generative Adversarial Networks (GANs), SSIM loss was combined with other loss functions to achieve better results. The combination of SSIM with traditional pixel-wise losses like L1 or L2 helped in preserving the structural integrity of the super-resolved images, leading to improvements in both SSIM and Peak Signal-to-Noise Ratio (PSNR) metrics. This approach was particularly effective in maintaining fine details and textures in the super-resolved images, demonstrating the advantages of using SSIM in conjunction with other loss functions

2.9.2 MS-SSIM Loss

Overview

The MS-SSIM loss is employed to enhance the perceptual quality of the translated images by focusing on structural similarity at multiple scales. This loss function evaluates the similarity between the generated image and the real image based on luminance, contrast, and structure, considering these aspects at different resolutions to capture fine details as well as global structures.

MS-SSIM Calculation

MS-SSIM Calculation The Structural Similarity Index (SSIM) compares local patterns of pixel intensities that have been normalized for luminance and contrast. The MS-SSIM extends this concept by computing the SSIM over multiple scales. The formula for SSIM between two patches x and y is:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

where:

- μ_x and μ_y are the mean intensities of patches x and y .
- σ_x^2 and σ_y^2 are the variances of patches x and y .
- σ_{xy} is the covariance between patches x and y .
- C_1 and C_2 are constants to stabilize the division with weak denominator values.

For MS-SSIM, the image is repeatedly downsampled, and SSIM is calculated at each scale j :

$$\text{MS-SSIM}(X, Y) = \left[\prod_{j=1}^M \text{SSIM}_j(X, Y) \right]^{1/M}$$

where M is the number of scales.

MS-SSIM Loss

The MS-SSIM loss function is utilized in the context of optimizing machine learning models, particularly in image reconstruction and enhancement tasks. It is defined as:

$$\mathcal{L}_{\text{MS-SSIM}} = 1 - \text{MS-SSIM}(\mathbf{x}, \mathbf{y}),$$

where $\mathbf{x}$ represents the ground truth image patch and $\mathbf{y}$ denotes the predicted image patch. This loss function aims to maximize the MS-SSIM index, thereby ensuring that the predicted image maintains structural similarity to the ground truth image.

Usage in image to image translation

The Multi-Scale Structural Similarity Index (MS-SSIM) loss function has been effectively utilized in various image-to-image translation tasks to enhance the perceptual quality and structural integrity of generated images. For instance, in the "Detail-Preserving CycleGAN-AdaIN Framework"[32] study, MS-SSIM loss was integrated into the reconstruction loss function to improve image quality in the translation from photographs to ink paintings. This integration led to significant improvements in metrics like Fréchet Inception Distance (FID), Kernel Maximum Mean Discrepancy (MMD), Peak Signal-to-Noise Ratio (PSNR), and SSIM compared to existing methods. Another study on "Loss Functions for Image Restoration with Neural Networks"[34] explored various loss functions, including MS-SSIM, for tasks such as denoising and demosaicking. The use of MS-SSIM helped reduce artifacts and enhanced the overall perceptual quality by preserving local structures more effectively than single-scale SSIM. Furthermore, the "HistoHDR-Net for HDR Image Translation"[2] research employed MS-SSIM as part of its loss function to improve the structural similarity of translated HDR images. This approach resulted in better representation of details and higher scores in perceptual image quality evaluation metrics

compared to simple SSIM. Additionally, in super-resolution tasks using Generative Adversarial Networks (GANs), MS-SSIM loss was combined with L1 loss, leading to superior results in both PSNR and SSIM metrics, thereby maintaining structural integrity across scales during the image enhancement process. Another significant example is presented in the paper "ADD-UNet: An Adjacent Dual-Decoder UNet for SAR-to-Optical Translation"[18]. This study introduces the ADD-UNet architecture, which enhances the structural features, details, and edge sharpness of generated images. The proposed method employs a hybrid loss function that combines the strengths of cGAN loss, L1 loss, and MS-SSIM loss. This combination proves to be more effective than using individual loss functions alone, as it leverages the detailed feature preservation capability of MS-SSIM and the luminance and color consistency maintained by L1 loss. Another work done by Tasnim Ahmed [1] usage MS-SSIM alognside with ResNet-9 architecture to enhance paired PET to CT image. The experimental results demonstrate that the hybrid loss function significantly improves the quality of translated images, achieving superior results in SAR-to-optical image translation tasks. These studies collectively highlight the effectiveness of incorporating MS-SSIM loss in image-to-image translation frameworks, demonstrating its role in producing high-quality, structurally coherent images across various applications.

2.9.3 Perceptual-Loss

Overview

Perceptual loss, also known as feature reconstruction loss, uses a pre-trained deep neural network (typically a classification network like VGG-16) to measure the perceptual similarity between the generated image and the real image. This loss function captures high-level features and textures that are crucial for perceptual quality.

Perceptual Loss Calculation

The perceptual loss is calculated by comparing feature representations of the generated and real images extracted from a pre-trained network. For a pre-trained network ϕ , the perceptual loss between a generated image G and a real image Y is defined as:

$$\mathcal{L}_{\text{perc}} = \sum_i \lambda_i \|\phi_i(G) - \phi_i(Y)\|_2^2$$

where:

- ϕ_i represents the activations of the i -th layer of the pre-trained network.
- λ_i is a weighting factor for the i -th layer.
- $\|\cdot\|_2$ denotes the Euclidean distance

Usage in Image to Image Translation

Perceptual loss, which leverages deep features extracted from pretrained networks, has been extensively utilized in image-to-image translation tasks to improve the perceptual quality and fidelity of generated images. In the influential work by Johnson et al. (2016) titled "Perceptual Losses for Real-Time Style Transfer and Super-Resolution,"[11] the authors introduced perceptual loss functions based on high-level feature maps from VGG networks. This approach demonstrated substantial improvements in maintaining the perceptual quality of images during style transfer and super-resolution tasks compared to traditional pixel-wise loss functions. Another notable application is found in the "Pix2Pix" framework by Isola et al. (2017), [10] where perceptual loss was employed to enhance the realism of image-to-image translation tasks. By using a discriminator network to compute the perceptual loss, the Pix2Pix model achieved impressive results in various tasks, such as generating photorealistic images from sketches and translating aerial images to maps. The study "High-Resolution Image Synthesis and Semantic Manipulation with Conditional GANs" by Wang et al. (2018) [26] also incorporated perceptual loss to improve the fidelity of high-resolution image synthesis. By utilizing deep features from a pre-trained network, the model achieved high-quality results in tasks like image synthesis and semantic manipulation, demonstrating the effectiveness of perceptual loss in preserving detailed textures and structures. In the domain of super-resolution, the work "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network" by Ledig et al. (2017)[14] introduced the SRGAN model, which used a combination of perceptual loss, adversarial loss, and content loss. The inclusion of perceptual loss, computed from the feature maps of a VGG network, significantly improved the perceptual quality of super-resolved images, making them more visually appealing and closer to human perception. Additionally, in the study "Unsupervised Image-to-Image Translation Networks" by Liu et al. (2017)[16], perceptual loss was used to guide the training of unsupervised translation models. This approach helped in achieving high-quality results in tasks such as object transfiguration and season transfer, by ensuring that the translated images retained essential structural and content details from the input images. Overall, these studies highlight the versatility and effectiveness of perceptual loss in various image-to-

image translation and enhancement tasks. By focusing on high-level features from pretrained networks, perceptual loss functions help produce images that are not only quantitatively superior but also more visually coherent and appealing, aligning closely with human visual perception.

2.9.4 Variational Autoencoder Loss

Introduction

Variational Autoencoders (VAEs) are a type of generative model that combine neural networks with variational inference. VAEs aim to learn a probabilistic model of the data in a latent space, allowing for the generation of new data points. The VAE loss function is crucial for training these models, comprising two main components: the reconstruction loss and the Kullback-Leibler (KL) divergence.

VAE: Definition and Mathematical Formulation

A VAE consists of two neural networks: the encoder, which maps input data $\mathbf{x}$ to a latent representation $\mathbf{z}$, and the decoder, which reconstructs the input data from the latent representation. The encoder produces parameters of the approximate posterior distribution $q_\phi(\mathbf{z}|\mathbf{x})$, typically a Gaussian distribution with mean $\mu_\phi(\mathbf{x})$ and variance $\sigma_\phi^2(\mathbf{x})$. The decoder defines the likelihood $p_\theta(\mathbf{x}|\mathbf{z})$.

Reconstruction Loss The reconstruction loss measures how well the VAE can reconstruct the input data $\mathbf{x}$ from the latent representation $\mathbf{z}$. It is usually defined as the negative log-likelihood of the input data given the latent representation:

$$\mathcal{L}_{\text{reconstruction}} = -\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})].$$

For continuous data with Gaussian likelihood, this can be written as:

$$\mathcal{L}_{\text{reconstruction}} = \frac{1}{2\sigma^2} \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \frac{d}{2} \log(2\pi\sigma^2),$$

where $\hat{\mathbf{x}} = \mu_\theta(\mathbf{z})$ is the reconstruction of $\mathbf{x}$, d is the dimensionality of the data, and σ^2 is the variance of the likelihood model, often assumed to be 1.

KL Divergence

The KL divergence term regularizes the encoder by ensuring that the approximate posterior $q_\phi(\mathbf{z}|\mathbf{x})$ is close to the prior $p(\mathbf{z})$. The prior is typically chosen to be a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The KL divergence is given by:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})).$$

For a Gaussian approximate posterior $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_\phi(\mathbf{x}), \sigma_\phi^2(\mathbf{x}))$, this becomes:

$$\mathcal{L}_{\text{KL}} = \frac{1}{2} \sum_{j=1}^d \left(1 + \log(\sigma_{\phi,j}^2(\mathbf{x})) - \mu_{\phi,j}^2(\mathbf{x}) - \sigma_{\phi,j}^2(\mathbf{x}) \right),$$

where d is the dimensionality of the latent space.

VAE Loss Function

The total VAE loss function combines the reconstruction loss and the KL divergence:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [-\log p_\theta(\mathbf{x}|\mathbf{z})] + D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})).$$

This objective encourages the model to accurately reconstruct the input data while regularizing the latent space to follow the prior distribution.

Here are some research papers that have utilized the Structural Similarity Index (SSIM) metric for image generation:

Usage in image to image translation

Variational autoencoders (VAEs) have been extensively used in image-to-image translation tasks, with various papers exploring their applications and effectiveness. For instance, the work by Kingma and Welling (2014) introduced the foundational concepts of VAEs and has been referenced widely, including in image translation contexts. In "Unsupervised Image-to-Image Translation Networks" [16] by Liu et al.

(2017), VAEs are employed to learn disentangled representations, facilitating better translation between image domains. Similarly, "Diverse Image-to-Image Translation via Disentangled Representations" by Lee et al. [15] uses a VAE-GAN architecture to enhance the diversity of translated images. Zhu et al. (2017) in "Toward Multimodal Image-to-Image Translation"[37] leverage a VAE framework to enable multimodal outputs. Other notable works include "Adversarially Learned Inference"[5] by Dumoulin et al. (2016), which combines VAEs with GANs for improved translation results. These references highlight the significant role of VAE losses in advancing the field of image-to-image translation.

Chapter 3

Proposed Methodology

The proposed methodology aims to address the research objective of accurately translating PET images into corresponding CT images using four distinct architectures. These architectures incorporate modifications on two different baseline models - cycleGAN & CUT

The research design for this study involves the exploration of four architectures:

3.1 First Architecture

3.1.1 Overview

The model is designed to translate PET images to CT images using an unpaired image-to-image translation approach, incorporating spatial attention mechanisms to enhance the translation quality. The architecture is based on a modified CycleGAN 2.4 framework with specific enhancements to handle the unique characteristics of PET and CT images.

Model Components

Generator Architecture (GAB and GBA)

Downsampling The input image 256x256 (either PET or CT) is passed through several convolutional layers that reduce the spatial dimensions. This process extracts high-level features from the image, which are essential for accurate translation.

Spatial Attention After downsampling, spatial attention mechanisms are applied to the feature maps. Spatial attention helps the model to focus on important regions of the image, enhancing the translation process by highlighting significant features and suppressing irrelevant ones.

ResNet9 Blocks The downsampled and attention-enhanced feature maps are then processed through ResNet9 blocks. These blocks contain residual connections that facilitate the training of deep networks by allowing gradients to flow more easily through the network.

Upsampling The processed feature maps are then upsampled using transposed convolutions or other upsampling techniques to reconstruct the image in the target domain. This step increases the spatial dimensions back to the original image size.

Discriminator Architecture

Discriminator for Domain A (CT Images) The discriminator (70*70 patch-gan) for domain A (CT images) is designed to distinguish between real CT images and fake CT images generated by the generator. It consists of several convolutional layers followed by fully connected layers that output a probability score indicating the likelihood that the input image is real or fake.

Discriminator for Domain B (PET Images) Similarly, the discriminator for domain B (PET images) distinguishes between real PET images and fake PET images reconstructed from CT images.

Loss Functions

Cycle Consistency Loss Ensures that an image translated to the target domain and then back to the source domain remains consistent with the original image. This loss helps in preserving the content and structure of the original image during the translation process.

Adversarial Loss Used to train the GAN framework, this loss ensures that the generator produces realistic images that can fool the discriminator. The adversarial loss is computed separately for both domains (PET to CT and CT to PET).

Identity Loss Identity loss is an additional constraint used in CycleGAN to preserve the color and structure of the input images when the input image is already

in the target domain. This loss encourages the generator to be an identity mapping when real samples of the target domain are provided as input.

3.1.2 Training Process

Forward Cycle (PET to CT to PET)

- The real PET image from domain B is input into the generator G_{BA} , which translates it into a fake CT image in domain A.
- The fake CT image is then passed through the generator G_{AB} , which reconstructs it back into a PET image.
- The cycle consistency loss is computed between the original PET image and the reconstructed PET image.

Backward Cycle (CT to PET to CT)

- Similarly, the real CT image from domain A is input into the generator G_{AB} , which translates it into a fake PET image in domain B.
- The fake PET image is then passed through the generator G_{BA} , which reconstructs it back into a CT image.
- The cycle consistency loss is computed between the original CT image and the reconstructed CT image.

Discriminator Training The discriminators for both domains are trained to distinguish between real images and fake images generated by the corresponding generators.

3.1.3 Spatial Attention Mechanism

Attention Map Generation

The spatial attention mechanism generates an attention map, which assigns different weights to different spatial regions of the feature map. This is typically done by computing the attention map using a series of convolutional operations followed by a sigmoid activation function.

The attention map $\mathbf{M}$ can be defined as:

$$\mathbf{M} = \sigma(\mathbf{F}_{\text{conv}}(\mathbf{F}_{\text{in}}))$$

where:

- $\mathbf{F}_{\text{in}}$ is the input feature map
- $\mathbf{F}_{\text{conv}}$ represents convolutional operations applied to the input feature map.
- σ is the sigmoid activation function, which normalizes the output to the range $[0, 1]$.

Applying the Attention Map

Once the attention map $\mathbf{M}$ is generated, it is used to modulate the input feature map $\mathbf{F}_{\text{in}}$ to produce an enhanced feature map $\mathbf{F}_{\text{out}}$. This is done through element-wise multiplication:

$$\mathbf{F}_{\text{out}} = \mathbf{M} \odot \mathbf{F}_{\text{in}}$$

where:

- $\mathbf{F}_{\text{out}}$ is the output feature map with enhanced focus on significant regions.
- $\odot$ denotes element-wise multiplication.

Integration into the Network

In the proposed architecture, the spatial attention mechanism is integrated after the downsampling stage. This ensures that the attention mechanism operates on a compact representation of the input image, making it computationally efficient and effective in highlighting important features.

Mathematical Formulation Let's delve deeper into the mathematical formulation of the spatial attention mechanism:

- **Input Feature Map:** Let $\mathbf{F}_{\text{in}} \in \mathbb{R}^{C \times H \times W}$ be the input feature map, where C is the number of channels, and H and W are the height and width of the feature map, respectively.
- **Convolutional Operations:** The convolutional operations $\mathbf{F}_{\text{conv}}$ can be represented as a series of convolutions with learnable parameters $\mathbf{W}_{\text{conv}}$:

$$\mathbf{F}_{\text{conv}}(\mathbf{F}_{\text{in}}) = \mathbf{W}_{\text{conv}} * \mathbf{F}_{\text{in}}$$

- **Attention Map:** The attention map $\mathbf{M}$ is obtained by applying the sigmoid

function to the convolutional output:

$$\mathbf{M} = \sigma(\mathbf{W}_{\text{conv}} * \mathbf{F}_{\text{in}})$$

- **Enhanced Feature Map:** The final enhanced feature map $\mathbf{F}_{\text{out}}$ is computed by element-wise multiplication of $\mathbf{M}$ and $\mathbf{F}_{\text{in}}$:

$$\mathbf{F}_{\text{out}} = \mathbf{M} \odot \mathbf{F}_{\text{in}}$$

Significance The incorporation of spatial attention mechanisms in the PET to CT image translation model has several significant benefits:

- **Enhanced Focus on Critical Regions:** Medical images often contain specific regions of interest that are crucial for diagnosis. Spatial attention helps the model to focus on these regions, improving the translation quality by emphasizing important features and structures.
- **Reduction of Irrelevant Information:** By suppressing irrelevant parts of the image, spatial attention reduces noise and enhances the clarity of the translated images. This is particularly important in medical imaging, where precision is paramount.
- **Improved Training Efficiency:** Attention mechanisms help in guiding the model to learn more effectively by highlighting informative regions. This can lead to faster convergence and better performance during training.
- **Robustness to Variations:** Spatial attention makes the model more robust to variations in the input images, such as differences in noise levels or artifacts. This robustness is essential for clinical applications where image quality can vary significantly.

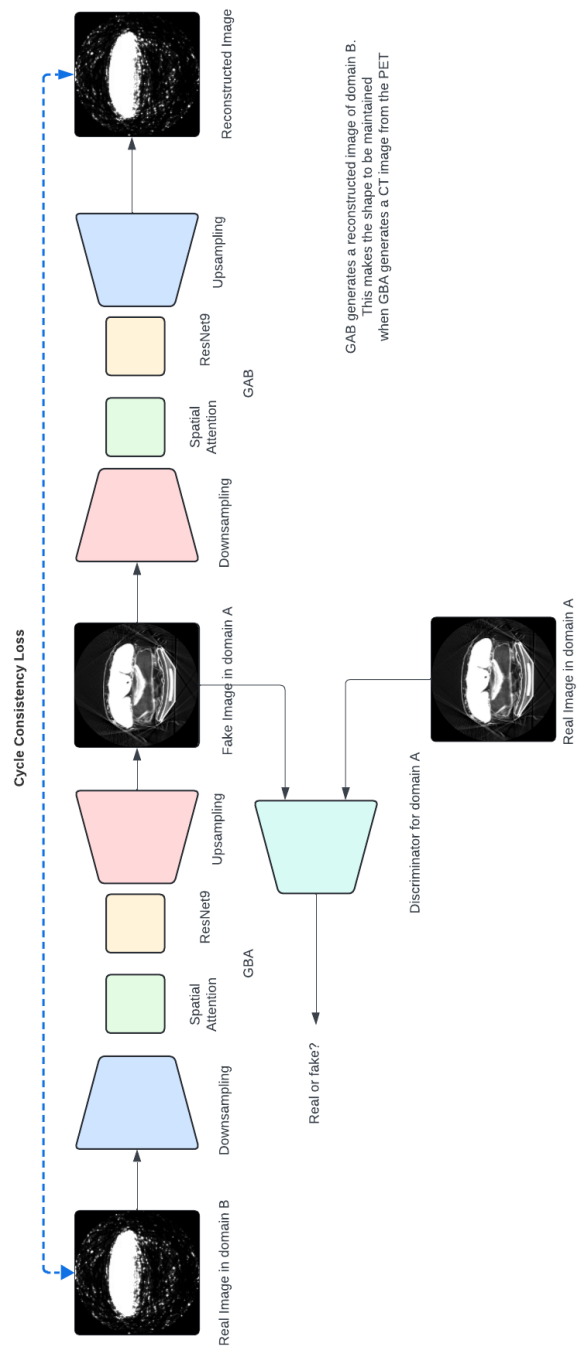


Figure 3.1: Visualization of the First Architecture

3.2 Second Architecture

The model translates PET images to CT images and vice versa, incorporating spatial attention mechanisms after every downsampling layer in the generator architecture. This setup aims to enhance the focus on significant regions of the image at multiple scales, improving the translation quality and accuracy. The architecture is based on a CycleGAN framework with specific modifications to handle the unique characteristics of PET and CT images.

Apart from the positioning of the spatial attention mechanism other modules remains the same

3.2.1 Spatial Attention Mechanism

Attention Map Generation After Each Downsampling Layer

The spatial attention mechanism generates attention maps after each downsampling layer. These maps assign different importance to different spatial regions of the feature map, helping the model to focus on significant regions at multiple scales.

The attention map $\mathbf{M}_i$ for the i -th downsampling layer is defined as:

$$\mathbf{M}_i = \sigma(\mathbf{F}_{\text{conv},i}(\mathbf{F}_{\text{in},i}))$$

where:

- $\mathbf{F}_{\text{in},i}$ is the input feature map for the i -th downsampling layer.
- $\mathbf{F}_{\text{conv},i}$ represents convolutional operations applied to $\mathbf{F}_{\text{in},i}$.
- σ is the sigmoid activation function, which normalizes the output to the range $[0, 1]$.

Applying the Attention Map After Each Downsampling Layer

The attention map $\mathbf{M}_i$ is used to modulate the input feature map $\mathbf{F}_{\text{in},i}$ through element-wise multiplication to produce an enhanced feature map $\mathbf{F}_{\text{out},i}$:

$$\mathbf{F}_{\text{out},i} = \mathbf{M}_i \odot \mathbf{F}_{\text{in},i}$$

where:

- $\mathbf{F}_{\text{out},i}$ is the output feature map with enhanced focus on significant regions for the i -th downsampling layer.
- $\odot$ denotes element-wise multiplication.

Integration into the Network

In the model, the spatial attention mechanism is integrated after each downsampling layer in the generator architecture (both GBA and GAB). This ensures that the attention mechanism operates on progressively compact representations of the input image, making it computationally efficient and effective in highlighting important features at multiple scales.

Mathematical Formulation Let's delve deeper into the mathematical formulation of the spatial attention mechanism:

- - Input Feature Map: Let $\mathbf{F}_{\text{in},i} \in \mathbb{R}^{C \times H \times W}$ be the input feature map for the i -th downsampling layer, where C is the number of channels, and H and W are the height and width of the feature map, respectively.
- Convolutional Operations: The convolutional operations $\mathbf{F}_{\text{conv},i}$ for the i -th downsampling layer can be represented as a series of convolutions with learnable parameters $\mathbf{W}_{\text{conv},i}$:

$$\mathbf{F}_{\text{conv},i}(\mathbf{F}_{\text{in},i}) = \mathbf{W}_{\text{conv},i} * \mathbf{F}_{\text{in},i}$$

- Attention Map: The attention map $\mathbf{M}_i$ for the i -th downsampling layer is obtained by applying the sigmoid function to the convolutional output:

$$\mathbf{M}_i = \sigma(\mathbf{W}_{\text{conv},i} * \mathbf{F}_{\text{in},i})$$

- Enhanced Feature Map: The final enhanced feature map $\mathbf{F}_{\text{out},i}$ for the i -th downsampling layer is computed by element-wise multiplication of $\mathbf{M}_i$ and $\mathbf{F}_{\text{in},i}$:

$$\mathbf{F}_{\text{out},i} = \mathbf{M}_i \odot \mathbf{F}_{\text{in},i}$$

Significance

The incorporation of spatial attention mechanisms after each downsampling layer in the PET to CT image translation model has several significant benefits:

- **Enhanced Focus on Critical Regions at Multiple Scales:** Medical images contain specific regions of interest that vary in size and importance. Applying spatial attention after each downsampling layer helps the model to focus on these regions at multiple scales, improving the translation quality by emphasizing important features and structures.

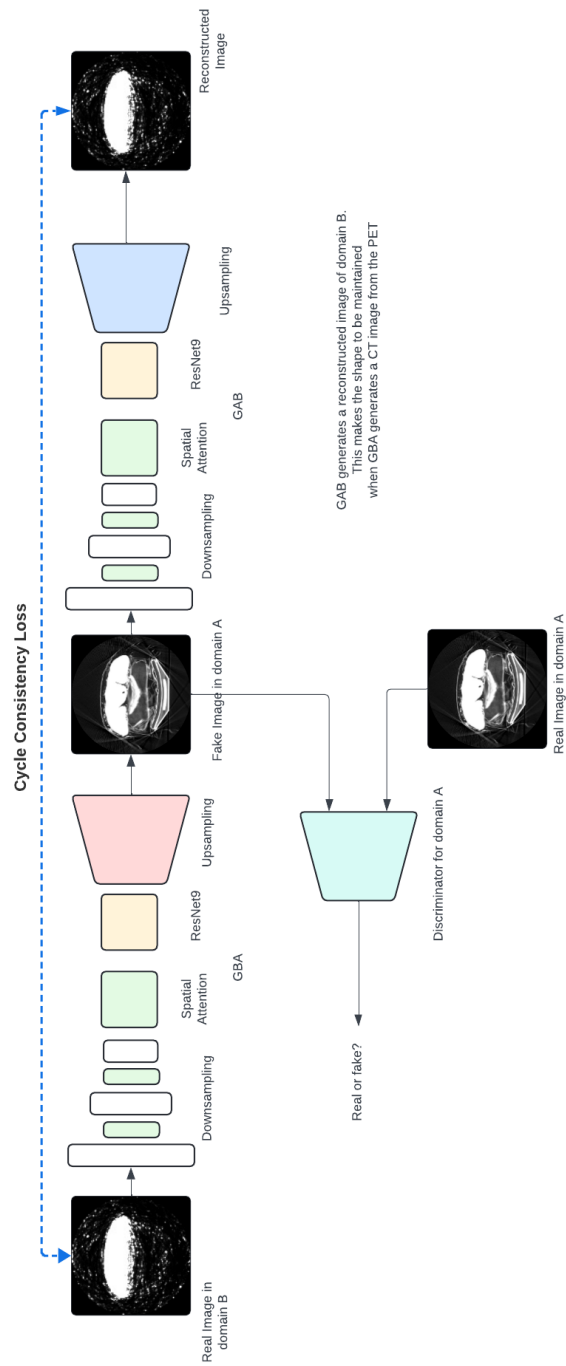


Figure 3.2: Visualization of the Second Architecture

3.3 Third Architecture

The updated model enhances the unpaired PET to CT image translation by incorporating spatial attention mechanisms, MS-SSIM loss, and perceptual loss. Spatial attention is applied after downsampling layers in the generator architecture, focusing on significant regions of the image. MS-SSIM loss functions are used to Capturing Multi-Scale Structural Information to enhance perceptual quality, VGG-16 pretrained is used to Capture High-Level Features to improve visual quality. This setup aims to improve the quality and accuracy of the translated images by leveraging the combined benefits of these mechanisms and loss functions.

This architecture remains same as architecture 1 apart from the additional MS-SSIM & perceptual loss.

3.3.1 Multi-Scale Structural Similarity Index (MS-SSIM) Loss

MS-SSIM overview & calculation

MS-SSIM overview is presented in 2.9.3 & calculation is presented in details in 2.9.3

Integration into the Network

In the network, the MS-SSIM loss is computed between the generated image and the reconstructed image. This loss is then backpropagated to update the weights of the generator, ensuring that the generated images have high structural similarity to the real images at multiple scales.

Significance

- **Capturing Multi-Scale Structural Information:** MS-SSIM ensures that the generated images preserve structural information at different scales, making them perceptually similar to real images.
- **Enhanced Perceptual Quality:** By focusing on structural similarity, the MS-SSIM loss helps in generating images with better perceptual quality, which is crucial for medical imaging applications.
- **Robustness to Artifacts:** The multi-scale approach makes the model robust to artifacts, ensuring that both local and global structures are preserved.

3.3.2 Perceptual Loss

Perceptual Loss Overview & Calculation

Perceptual loss overview is presented in 2.9.3 & calculation is presented in details in 2.9.3

Integration to the network

- **Feature Extraction:** The generated image G and the reconstructed image G' are passed through the pre-trained VGG-16 network.
- **Feature Comparison:** The activations from several intermediate layers (typically convolutional layers) are extracted and compared using the Euclidean distance.
- **Loss Backpropagation:** The perceptual loss is backpropagated through the generator to update its weights, ensuring that the generated images closely match the real images in terms of high-level features and textures.

Significance

- **Capturing High-Level Features:** Perceptual loss captures high-level semantic information and textures, which are often missed by pixel-wise losses.
- **Improved Visual Quality:** By focusing on perceptual similarity, this loss improves the visual quality of the generated images, making them more realistic and detailed.
- **Effective for Medical Imaging:** In medical imaging, capturing fine details and textures is crucial. Perceptual loss ensures that these aspects are well-preserved in the translated images.

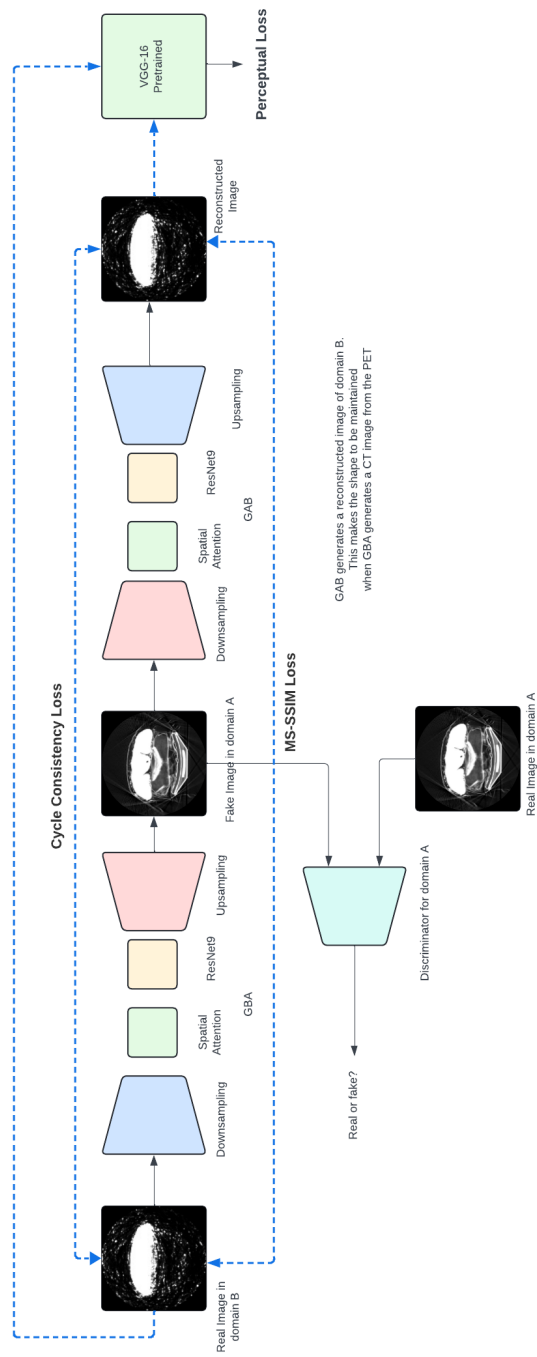


Figure 3.3: Visualization of the Third Architecture

3.4 Fourth Architecture

This model is based on the Contrastive Unpaired Translation (CUT) framework which is presented in details in 2.5, which enhances the translation quality by leveraging self-attention mechanisms, multilayer, patchwise contrastive loss and using hinge loss objective instead of lsgan. The self-attention mechanism is applied after downsampling layers to focus on important features within the image, and the multilayer, patchwise contrastive loss ensures that patches from the generated image closely match patches from the real image, improving the overall consistency and realism of the translated outputs.

3.4.1 Self-Attention Mechanism

Self-Attention Overview

Self-attention mechanisms allow the model to weigh the importance of different parts of the input feature map, dynamically focusing on the most relevant features. This is particularly useful in medical image translation, where specific regions contain critical information.

Self-Attention Calculation

Given an input feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, the self-attention mechanism computes a weighted sum of all positions in the feature map: $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$ where:

- $Q = W_Q\mathbf{F}$, $K = W_K\mathbf{F}$, and $V = W_V\mathbf{F}$ are the query, key, and value matrices, respectively.
- W_Q , W_K , and W_V are learnable weight matrices.
- d_k is the dimension of the key vectors.

Integration into the Network

The self-attention mechanism is integrated after the downsampling layers. This helps the network to dynamically focus on relevant features at lower resolutions, improving the feature representation before upsampling:

$$\mathbf{F}_{\text{att}} = \text{Attention} (W_Q\mathbf{F}, W_K\mathbf{F}, W_V\mathbf{F})$$

Significance

Enhanced Focus on Relevant Features Self-attention allows the model to dynamically focus on the most relevant parts of the image, improving feature extraction and translation quality.

3.4.2 Hinge Loss Objective

Overview

In this enhanced CUT framework, the hinge loss objective is used instead of the least squares GAN (LSGAN) loss for training the discriminator. The hinge loss objective has been shown to improve the stability and performance of GANs, especially in image synthesis tasks. This loss helps in better separating real and fake samples, leading to more realistic generated images.

Hinge Loss Calculation

The hinge loss objective for GANs is defined as follows:

Discriminator Loss The discriminator D aims to maximize the margin between real and fake samples. For a real image x and a generated (fake) image $G(z)$:

$$\mathcal{L}_D = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\max(0, 1 - D(x))] + \mathbb{E}_{z \sim p_z(z)}[\max(0, 1 + D(G(z)))]$$

where:

- $p_{\text{data}}(x)$ is the distribution of real images.
- $p_z(z)$ is the distribution of the input noise z .
- $D(x)$ represents the discriminator's output for real images.
- $D(G(z))$ represents the discriminator's output for generated images.

Generator Loss The generator G aims to produce images that can fool the discriminator. The hinge loss for the generator is defined as:

$$\mathcal{L}_G = -\mathbb{E}_{z \sim p_z(z)}[D(G(z))]$$

This encourages the generator to produce images that the discriminator classifies as real.

Integration into the CUT Framework

- **Discriminator Training:** The discriminator is trained to maximize the hinge loss objective. This involves minimizing the loss for real images when $D(x)$ is less than 1 and minimizing the loss for fake images when $D(G(z))$ is greater than -1. This creates a margin of separation between real and fake samples, leading to better discrimination.
- **Generator Training:** The generator is trained to minimize the negative output of the discriminator for generated images. This means the generator aims to produce images that receive high scores from the discriminator, pushing the discriminator's output towards positive values, indicating real images.

Significance

- **Sharper and More Realistic Images:** Hinge loss has been shown to produce sharper and more realistic images by encouraging a clearer margin between real and fake samples.
- **Better Separation of Real and Fake Samples:** By maximizing the margin between real and fake images, hinge loss improves the discriminator's ability to distinguish between real and fake samples, enhancing the overall quality of the generated images.

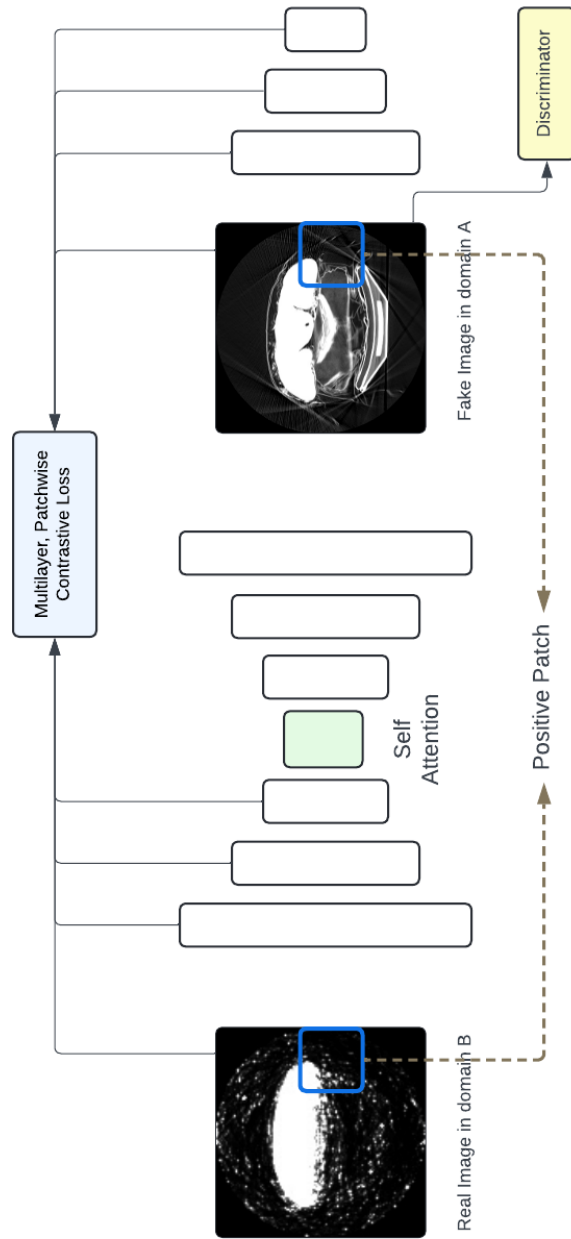


Figure 3.4: Visualization of the Fourth Architecture

3.5 Evaluation Metrics

After training the model, the following evaluation metrics will be employed to assess the performance of the translation:

- Structural Similarity Index (SSIM)
- Peak Signal-to-Noise Ratio (PSNR)

Chapter 4

Experimental Setup

4.1 Data Acquisition

We collect the data from the cancerimagearchive website. The dataset that we are using is named the **QIN-Breast QINBreast** Dataset.

This dataset comprises longitudinal PET/CT and quantitative MR images acquired to investigate treatment evaluation in breast cancer within the neoadjuvant context. The images were obtained at three specific time intervals: before treatment initiation (t1), after the first treatment cycle (t2), and either after the second treatment cycle or before surgery (t3). The PET/CT pictures were obtained using a custom-made support system that enabled the patient to lie face down, making it easier to align the images with the MRI data. The purpose of this collection is to offer clinical imaging data for the advancement and assessment of quantitative imaging techniques for early treatment evaluation in breast cancer therapy. The data is sourced from Vanderbilt University, under the supervision of Dr. Thomas E. Yankeelov. The PET/CT data were obtained using a GE Discovery STE scanner manufactured by GE Healthcare in Waukesha, WI, USA. A CT scan with a low milliamper-second (mAs) setting was obtained to adjust for attenuation in the emission data. The acquisition parameters for the transmission CT scan were as follows: the tube current was set at 80 mAs for a patient weighing 70 kg, and adjusted proportionally for all patients. The tube voltage was set at 120 KVp, and the pitch was 1.675/1. The amount of FDG administered was roughly 370 MBq (10 mCi) for a patient weighing 70 kg, and it was adjusted proportionally based on the patient's weight. The FDG was injected intravenously through a vein in the arm opposite to the afflicted breast. Emission data was gathered in 3D mode for 2 minutes per bed position after a duration of 60

minutes. The emission scan was initially obtained with the patient lying face down, focusing on the breast area. Subsequently, the scan was performed with the patient lying face up, covering the area from the skull to the middle of the femurs. Standard of care Images and studies in a supine position. photos were obtained in the prone position at time points t1 and t3, while only prone photos were obtained at t2.

4.2 Dataset Size

The dataset comprises 67 subjects. The images are categorized into three modalities: CT, PET, and MR. The initial dataset has a size of 11.41 gigabytes and consists of more than 100,000 DICOM pictures. We obtained a total of 110 CT and PET scans each for the 43 patients at three separate time stages, resulting in a sufficient dataset. Each scan consists of 83 slices, resulting in a total of 9130 pictures for both CT and PET. The final dataset comprises 18,260 pictures.

4.3 Dataset Preprocessing

The photos are initially presented in a paired configuration. The photos of each patient are stored in a dedicated folder, with a subfolder corresponding to the specific time stage. Every timestage folder consists of two additional subfolders labeled CT and PET. These two folders include a total of 83 related DICOM images. Initially, we convert the images into jpeg format and subsequently assign them random names to generate an unpaired dataset.

The CT scans have an initial resolution of 512x512 pixels, while the PET images have a resolution of 128x128 pixels. Initially, we employ bicubic interpolation to adjust their dimensions to 256x256 pixels. Next, we adhere to a conventional transformation process in which we convert the photos into pytorch tensors and normalize them by subtracting the mean value of 0.5 and dividing by the standard deviation of 0.5. Standardization ensures that pixel values are scaled to a specific range, which facilitates the training of deep learning models.

4.4 Model Implementation Details and Training

4.4.1 Dataset Split

We use an 80:20 train-test split for our QIN-BREAST dataset. Our models are trained using an unpaired dataset. However, we evaluate our models using a matched dataset in which the corresponding images have identical names and are stored in separate files for CT and PET scans. This provides us with the definitive and accurate information needed for the purpose of evaluation.

4.4.2 Performance Evaluation and Metric

A comprehensive range of metrics has been utilized to evaluate the efficacy of the CycleGAN model in translating PET data into CT images. These metrics analyze both the visual quality and quantitative correctness of the generated CT images.

Structural Similarity Index (SSIM)

The Structural Similarity Index (SSIM) is essential for assessing the perceptual quality of the translated CT images. This metric considers the structural information, brightness, and contrast of the images. Regarding the translation of PET to CT, a higher SSIM value signifies a superior retention of important characteristics and overall visual accuracy in the generated CT images compared to the ground truth (gt) CT images.

$$\text{SSIM}(CT_{\text{gt}}, CT_{\text{generated}}) = \frac{(2\mu_{\text{gt}}\mu_{\text{generated}} + c_1)(2\sigma_{\text{gt}, \text{generated}} + c_2)}{(\mu_{\text{gt}}^2 + \mu_{\text{generated}}^2 + c_1)(\sigma_{\text{gt}}^2 + \sigma_{\text{generated}}^2 + c_2)} \quad (4.1)$$

Peak Signal-to-Noise Ratio (PSNR)

The Peak Signal-to-Noise Ratio (PSNR) is used as a quantitative metric to assess the accuracy of the translated CT images. PSNR quantifies the level of noise and distortion in the generated pictures by evaluating the ratio between the highest potential power of the ground truth CT images and the power of translation errors.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (4.2)$$

Interpretation and Criteria for Success

When analyzing the results, a greater SSIM (Structural Similarity Index) and PSNR (Peak Signal-to-Noise Ratio) generally indicate superior performance in the conversion of PET pictures to CT images.

A higher SSIM value, approaching 1, signifies superior preservation of structural information, brightness, and contrast. This suggests that the generated CT pictures closely reflect the ground truth CT images. Similarly, a greater PSNR value signifies reduced noise and enhanced fidelity in the translated images.

The success criteria for each statistic should be established according to the precise objectives of our research. To assess the level of success in the translation process.

Limitations

Although these measures offer useful insights into the performance of the proposed models, it is crucial to recognize their limits, especially when applied to medical image translation from PET to CT.

Firstly, the presence of anatomical changes and disparities in imaging techniques between PET and CT can provide difficulties that are not completely measured by quantitative measurements. The intrinsic variability in the anatomy of patients and the methods by which contrast is produced may result in differences in specific areas, which might affect the accuracy of the translation.

Furthermore, the metrics primarily prioritize evaluating variations in individual pixels and similarities in structure, disregarding any perceptual distinctions that could be vital in the study of medical images. The perception of some qualities by human specialists may vary, highlighting the need to include clinical validation and expert opinion.

Moreover, the measures are highly responsive to interference and imperfections, and their effectiveness can be modified by the caliber of the training data and the intricacy of the translation assignment. When evaluating the usefulness of the translated images, it is important to carefully analyze the data and take into account the wider clinical context.

To summarize, although quantitative metrics offer significant quantitative assessments, it is essential to supplement them with qualitative evaluations and expert input to guarantee the trustworthiness and clinical significance of the translated PET to CT pictures.

4.4.3 Hyperparameters

Our experimentation involved a very limited exploration of hyperparameters as we had limited computational resources. Table-4.1 provides a detailed overview of the hyperparameters configured for each test run. Also apart from that, the batch size for all the run as 1, the learning rate was .00002, learning decay was set for half. To expedite training time while maintaining the essence of our analysis, the dataset size was reduced to 10% and the epoch was reduced to 50. For some of the models we trained on 100% data and the epoch was 100.

4.5 Testing

4.5.1 Hardware and Schedule

We have used the NVIDIA RTX 3090 GPU equipped with 24GB of VRAM and an Intel Core i9 Processor for conducting our experiments. The training time for 10% dataset and 50 epoch was around 4 hours & for 100% dataset and 100 epoch it was around 1.5-2 days. This shows even with such heavy setup our models took about couple of days to fully run, which is why it was portioned down to meet our deadline. 4.1 table shows the other parameters

Table 4.1: Parameters used for Testing

Baseline Model	Incorporation	Cycle	Identity	MSSSIM	SSIM	Perceptual
cycleGAN	None	15	0.5	-	-	-
cycleGAN	CBAM in Generator & Discriminator	15	0.5	-	-	-
cycleGAN	Self Attention Mechanism After Downsampling	15	0.5	-	-	-
cycleGAN	Spatial Attention Mechanism After Downsampling	15	0.5	-	-	-
cycleGAN	ResNet18 as backbone	15	0.5	-	-	-
cycleGAN	UNet256 as backbone	15	0.5	-	-	-
cycleGAN	Using style & content encoder separately	15	0.5	-	-	-
cycleGAN	Spatial Attention Mechanism with SSIM loss as a cycle-consistent loss	15	0.5	10	5	-
cycleGAN	Spatial Attention Mechanism with MSSSIM loss as a cycle-consistent loss	15	0.5	10	5	-
cycleGAN	Dual Attention Mechanism	15	0.5	-	-	-
cycleGAN	CBAM in Generator	15	0.5	-	-	-
cycleGAN	DUAL Attention Mechanism with CBAM	15	0.5	-	-	-
cycleGAN	Spatial Attention Mechanism with SSIM loss as a cycle-consistent loss & Perceptual Loss	15	0.5	10	5	-
cycleGAN	Spatial Attention Mechanism After Every Downsampling Layer	15	0.5	10	5	-
cycleGAN	ViT & ResNet as backbone	15	0.5	-	-	-
cycleGAN	ViT as backbone	15	0.5	-	-	-
CUT	None	N/A	N/A	N/A	N/A	N/A
CUT	Hinge Loss Objective	N/A	N/A	N/A	N/A	N/A
CUT	Spatial Attention Mechanism After Downsampling	N/A	N/A	N/A	N/A	N/A
CUT	Self Attention Mechanism After Downsampling	N/A	N/A	N/A	N/A	N/A
CUT	ViT & ResNet as backbone	N/A	N/A	N/A	N/A	N/A
CUT	ViT as backbone	N/A	N/A	N/A	N/A	N/A

Chapter 5

Result Analysis and Discussion

5.1 Experimental Results

The comparative performance of various GAN models and their modified counterparts is illustrated in Table 5.1. The metrics of interest include Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). In here there are multiple incorporation which we have tested are mentioned, alongside some popular unpaired models are evaluated to compare fairly. The results are shown for both the reduced version and the full version. But, for some incorporation and models we couldn't complete the full version's training due to resource constraints.

Table 5.1: Result of Different Unpaired Models with Various Incorporations

Baseline Model	Incorporation	Parameter Count	SSIM (10%)	PSNR (10%)	SSIM (100%)	PSNR (100%)
cycleGAN	None	28 Million	0.5126	15.73	0.518	16.06
cycleGAN	CBAM in Generator & Discriminator (SCAM-GAN)	28 Million	0.4955	14.2	-	-
cycleGAN	Self Attention Mechanism After Downsampling	28 Million	0.5121	13.62	-	-
cycleGAN	Spatial Attention Mechanism After Downsampling	28 Million	0.5623	15.48	0.5712	16.22
cycleGAN	Perturbation Consistency & Spatial Alignment (MSPC)	28 Million	0.4549	13.39	-	-
cycleGAN	ResNet18 as backbone	28 Million	0.4943	14.95	-	-
cycleGAN	UNet256 as backbone	28 Million	0.4495	14.3	-	-
cycleGAN	Using style & content encoder separately (MUNIT)	28 Million	0.5526	15.48	0.5588	16.72
cycleGAN	Spatial Attention Mechanism with SSIM loss as a cycle-consistent loss	28 Million	0.5198	13.49	0.5682	16.17
cycleGAN	Spatial Attention Mechanism with MSSSIM loss as a cycle-consistent loss	28 Million	0.5395	15.46	-	-
cycleGAN	Dual Attention Mechanism	28 Million	0.5457	15.69	-	-
cycleGAN	CBAM in Generator	28 Million	0.5112	13.52	-	-
cycleGAN	DUAL Attention Mechanism with CBAM	28 Million	0.5181	14.89	-	-
cycleGAN	Spatial Attention Mechanism with upscaled images	28 Million	0.5049	15.01	-	-
cycleGAN	Spatial Attention Mechanism with SSIM loss as a cycle-consistent loss & Perceptual Loss	28 Million	0.5341	15.72	0.5621	16.12
cycleGAN	Spatial Attention Mechanism with MS-SSIM loss as a cycle-consistent loss & Perceptual Loss	28 Million	0.5443	15.84	0.5722	16.52
cycleGAN	Spatial Attention Mechanism After Every Downsampling Layer	28 Million	0.5433	15.62	0.5714	16.32
cycleGAN	ViT & ResNet as backbone	208 Million	Crashed	Crashed	Crashed	Crashed
cycleGAN	ViT as backbone	184 Million	Crashed	Crashed	Crashed	Crashed
CUT	None	14 Million	0.5149	14.9	0.5248	15.32
CUT	Spatial Attention Mechanism After Downsampling	14 Million	0.5139	15	-	-
CUT	Self Attention Mechanism After Downsampling	14 Million	0.525	15.22	0.5325	15.84
CUT	Self Attention Mechanism After Downsampling with hinge loss objective	14 Million	0.5268	14.82	0.5811	16.38
CUT	ViT & ResNet as backbone	104 Million	0.5078	14.92	long time	long time
CUT	ViT as backbone	92 Million	0.4991	14.78	long time	long time

5.2 Comparison and Discussion

1. Spatial Attention Mechanism After Downsampling significantly outperforms the baseline CycleGAN and most other variants in terms of SSIM and PSNR, especially at 100% quality settings, showing the effectiveness of spatial attention mechanisms.
2. The Spatial Attention Mechanism with MS-SSIM and Perceptual Loss further improves the results, particularly in PSNR, indicating that combining different loss functions can lead to better image quality.
3. Spatial Attention Mechanism After Every Downsampling Layer also shows improved performance, indicating that adding attention layers at multiple stages can enhance the model's ability to capture fine details.
4. SCAM-GAN and ResNet18 as Backbone show poorer performance in terms of SSIM and PSNR compared to our incorporations.
5. UNet256 as Backbone performs worse than the baseline, suggesting that the chosen backbone architecture has a significant impact on performance.
6. Using Style & Content Encoder Separately (MUNIT) performs well but is still outperformed by our incorporations, especially in SSIM metrics at 100%.
7. The Self Attention Mechanism with Hinge Loss in CUT achieves the highest SSIM and PSNR at 100% quality settings among all CUT variants, indicating its effectiveness.
8. The Spatial Attention Mechanism After Downsampling also shows improved SSIM and PSNR over the baseline, confirming the utility of attention mechanisms.
9. The results indicate that attention mechanisms significantly enhance the performance of CUT models, even with fewer parameters compared to models with larger backbones like ViT & ResNet.
10. The ViT & ResNet as Backbone variant, despite having significantly more parameters (104 Million), does not outperform the green-marked incorporations in SSIM and PSNR, demonstrating the efficiency of attention mechanisms in improving model performance without drastically increasing model complexity.

11. Apart from the incorporation, MUNIT is the current state of the art architecture for this task. Our incorporation beat MUNIT in different scenarios

Chapter 6

Conclusion and Future Works

6.1 Conclusion

In the task of unpaired PET to CT image translation, various model incorporations have been tested and analyzed for their effectiveness in improving image quality. The incorporations primarily focused on adding attention mechanisms, leveraging advanced loss functions, and experimenting with different model backbones. The results show that our incorporations, specifically those involving spatial and self-attention mechanisms, have demonstrated significant improvements in both SSIM (Structural Similarity Index Measure) and PSNR (Peak Signal-to-Noise Ratio) metrics compared to baseline models and other tested variants.

Despite these improvements, the accuracy and quality of the translated images are not yet satisfactory for use in the medical domain. However, the enhancements achieved indicate a strong potential for further development and optimization. Our findings suggest that the tested incorporations outperform many current unpaired models, but there remains ample room for further exploration and refinement.

6.2 Future Work

To build upon the promising results obtained from the current study, several future works and potential incorporations can be explored. Each proposed incorporation is aimed at addressing specific limitations or enhancing particular aspects of the model:

6.2.1 Multi-scale Architecture for Generator and Discriminator

Rationale

Multi-scale architectures allow the model to capture features at different levels of granularity, which can be particularly beneficial for translating intricate details in medical images. Implementation: This incorporation will increase the parameter count, but the trade-off may be worthwhile for the potential gains in image quality and structural coherence.

6.2.2 Dual Discriminators

Introducing dual discriminators can help in balancing the generator and discriminator, potentially leading to more stable training and better quality images. Consideration: Care must be taken to ensure that the dual discriminator setup does not disrupt the generator-discriminator balance, which can be a delicate aspect of GAN training.

6.2.3 Using Ideas from CUT (Contrastive Unpaired Translation)

Rationale

CUT focuses on maximizing the mutual relationship between images using InfoNCE loss, which can be integrated with cycle-consistency loss to enhance image translation fidelity. This approach can help in preserving more structural details and improving the overall quality of the translated images.

6.2.4 Hyperparameter Tuning

Rationale

Systematic hyperparameter tuning can lead to significant improvements in model performance by finding the optimal settings for learning rate, batch size, and other critical parameters. Approach: Employing automated hyperparameter optimization techniques such as grid search or Bayesian optimization.

6.2.5 Dense ResNet Instead of ResNet-9

Rationale

Dense ResNets, with their more extensive connectivity, can enhance feature reuse and gradient flow, leading to potentially better performance than the simpler ResNet-9 architecture. Impact: This could lead to more accurate and high-quality image translations by leveraging deeper and more complex feature representations.

6.2.6 Attention UNET Instead of ResNet-9

Rationale

Incorporating attention mechanisms within a UNET architecture can help the model focus on the most relevant parts of the input images, thereby improving the quality of the output. Advantage: This modification is expected to enhance the model's ability to capture fine details, which is crucial for medical image translation.

6.2.7 Adding Variational Autoencoder Loss

Rationale

Introducing a variational autoencoder (VAE) loss can help in generating more diverse and realistic images by encouraging the model to learn a distribution over the latent space. Consideration: Care must be taken to ensure that this does not introduce irrelevant similarities between images, which could detract from the clinical utility of the generated CT images.

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